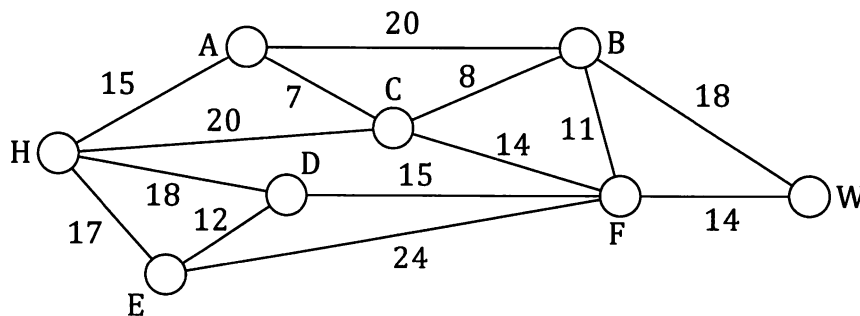


Chapter 6.

Shortest path.

Situation One.

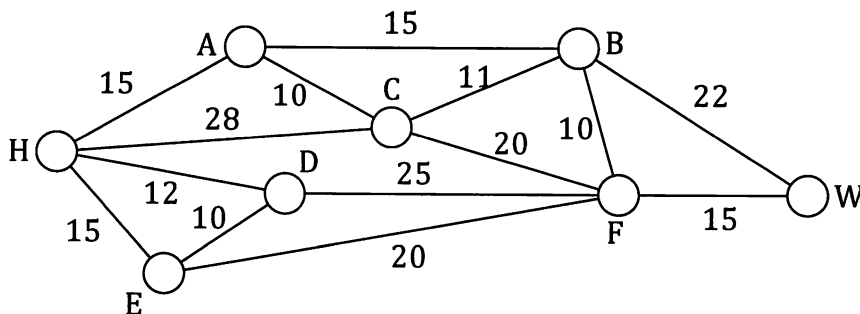
The diagram below shows a network of roads linking the locations, H, A, B, C, D, E, F and W shown as circles. (The circles are the vertices of the network, they are not loops. The reason for using open circles will become apparent later.) The diagram is not drawn to scale, and the roads are not necessarily straight, but the number on each road gives the road distance in kilometres between the locations (e.g. from H to A is 15 km).



A motorist wishes to travel from home, at H, to work, at W, by the shortest route. Which route should the motorist take and how long will it be?

Situation Two.

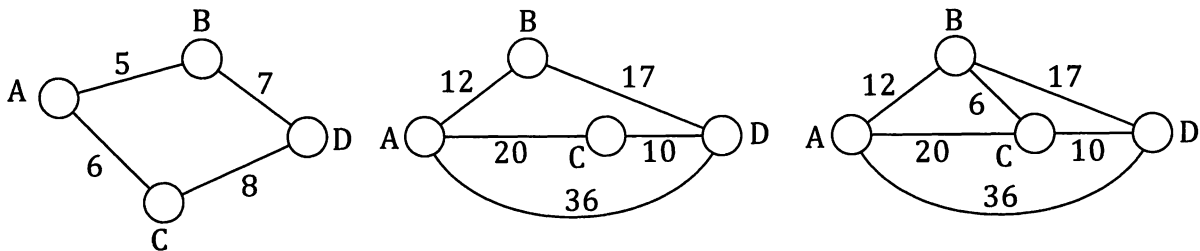
After travelling the shortest route a number of times the motorist of situation one realises that he wants to travel by the quickest route and, due to speed limits and road conditions, this may not be the shortest route. He uses his experience to estimate the time it takes him to travel each section of road in the network. These times, in minutes, are shown on the diagram below.



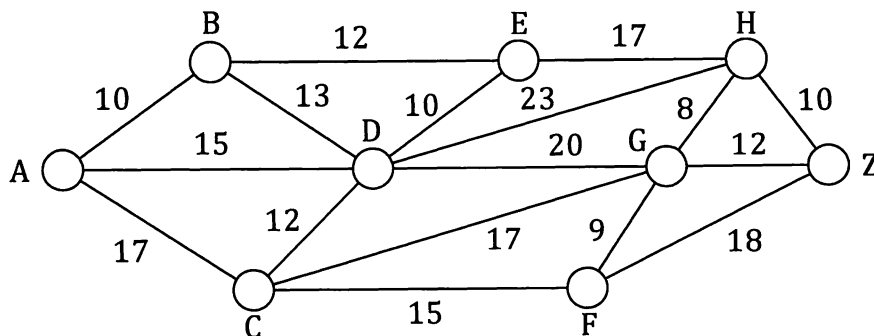
Assuming his estimates are correct which route should the motorist take to complete the journey from H to W as quickly as possible, and how many minutes will the journey take?

The situations on the previous page required us to find the **shortest path** from one vertex, point H, to another, point W. In situation one "shortest" meant least distance and in situation two "shortest" meant least time.

If the network is reasonably simple we may be able to identify the shortest path by inspection. For example you should be able to identify the shortest path from A to D in each of the following by inspection.



However, for a more complicated network this "inspection" method may not be easy. For example it would not be so easy to determine the shortest distance from A to Z in the network shown below, simply by inspection. We may arrive at an answer but can we be sure it is the **shortest** path?



For such cases we need a process, or **algorithm**, to follow that will enable us to find the shortest path. (In mathematics an *algorithm* is a method of solving a problem by repeating a sequence of steps.)

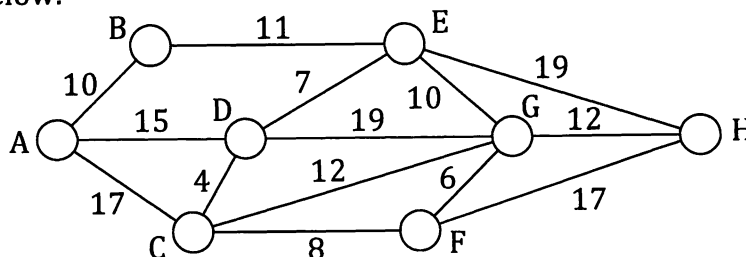
Shortest path – a systematic approach.

To determine the shortest path between two vertices, for networks in which this cannot easily be done by inspection, we proceed as shown in the example below.

Note: Previously we introduced the *length* of a path to mean the number of edges used, however, for weighted graphs it tends to be used to mean *total weight*.

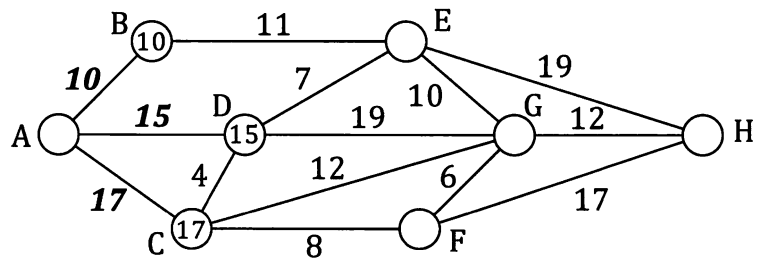
Example 1

Determine the length and the route of the shortest path from point A to point H in the network shown below.



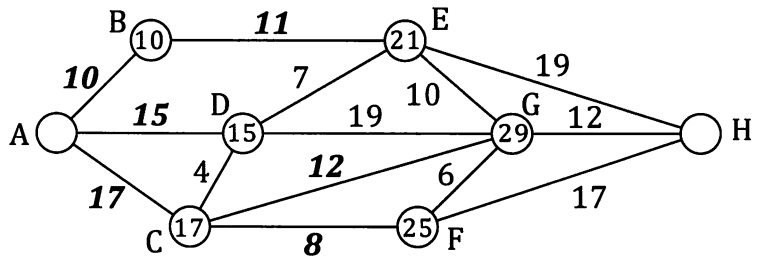
The systematic approach we will consider involves moving from A, towards H, placing a number at each vertex in turn, to indicate the length of the shortest path from A to that vertex.

From A the shortest path to B is of length 10 units, to D is 15 units and to C is 17 units. Note in the diagram on the right the placement of these numbers and also note that the numbers on the edges giving these shortest routes are now indicated in *bold italic*.



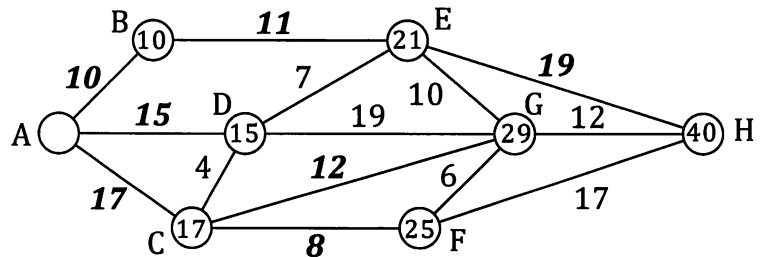
Note • Whilst indicating these edges in "bold italic" is reasonable for a book to do you are advised to circle the numbers on these edges using a different colour, say red, or perhaps to highlight the numbers with a highlighter pen.

If we continue this method to points E, F and G the diagram will then be as shown on the right.



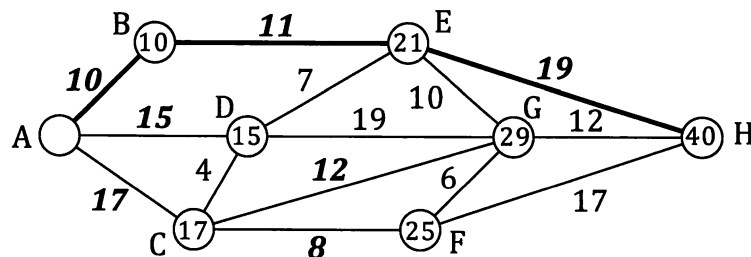
We can then complete the method by continuing on to point H:

Thus the shortest path from A to H is of length 40 units.



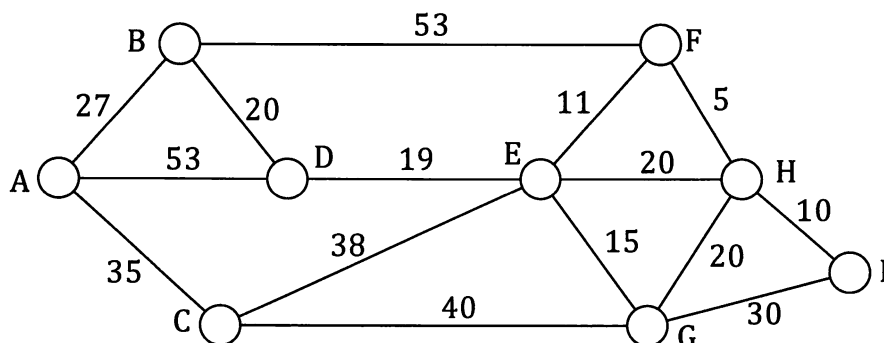
By backtracking from H back to A, along bold italic numbers only (or in your case red circled or highlighted numbers only) we see that the shortest path has come from H-E-B-A.

Thus the shortest path is 40 units and is achieved by the route ABEH:

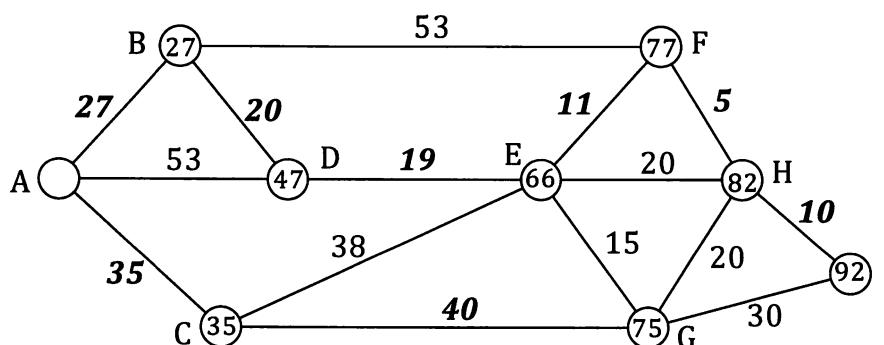


Example 2

The network shown below shows the time, in minutes, taken to travel along various sections of road. According to these times what is the quickest time that the journey from A to I could be completed in and what route would achieve this quickest journey?



Following the systematic approach the network becomes "marked up" as shown below.

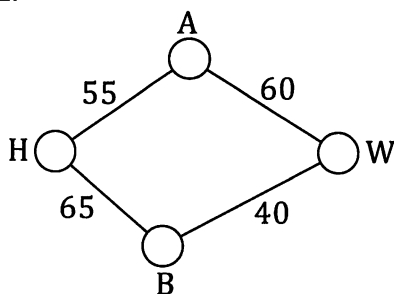


The shortest time for journeying from A to I is 92 minutes and is achieved by the route ABDEFHI.

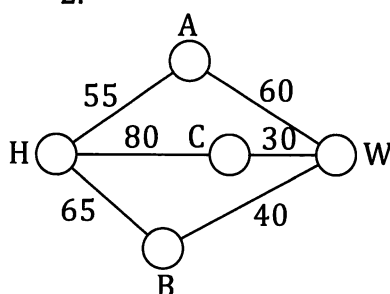
Exercise 6A

For networks 1 to 3 shown below determine, by inspection, the length (i.e. total weight) and route of the shortest path from H to W.

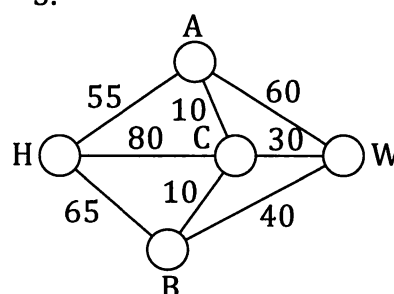
1.



2.

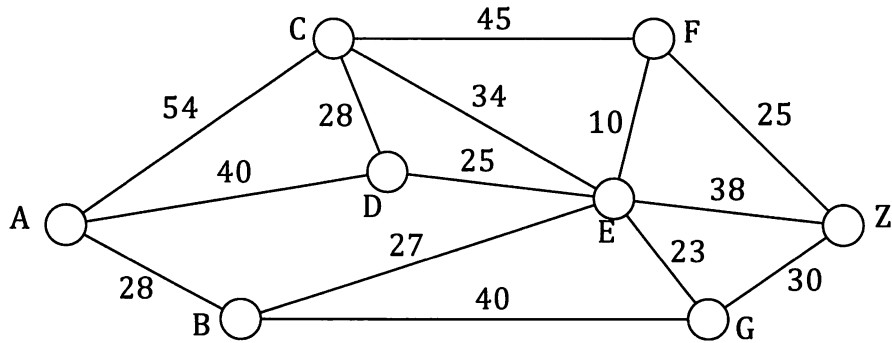


3.

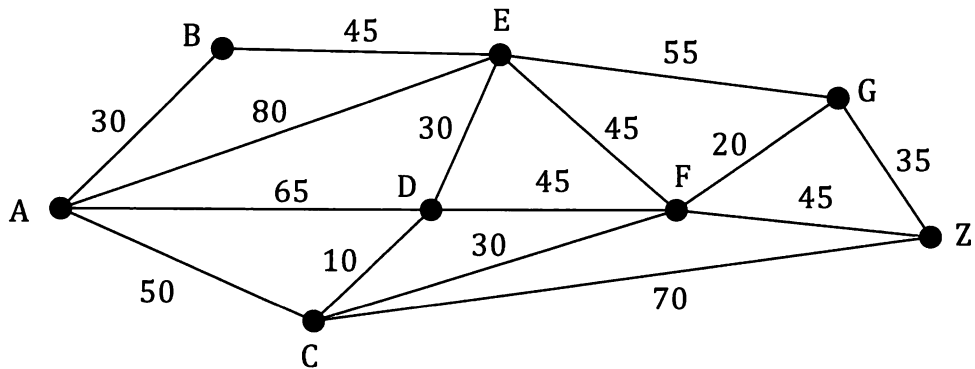


Use a systematic approach to determine the length (i.e. total weight) and the route of the shortest path from A to Z for the networks shown in questions 4 to 8.

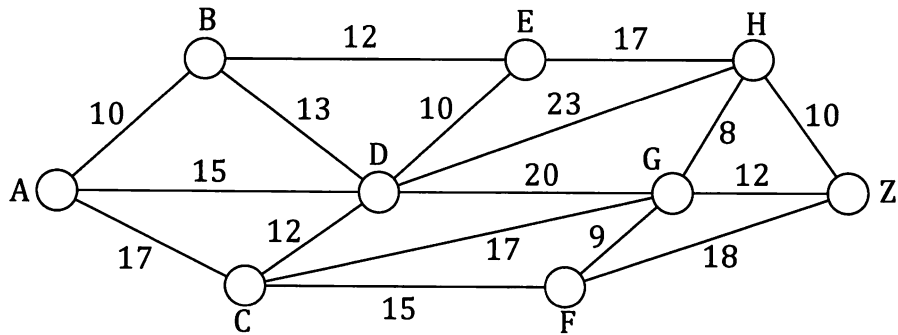
4.



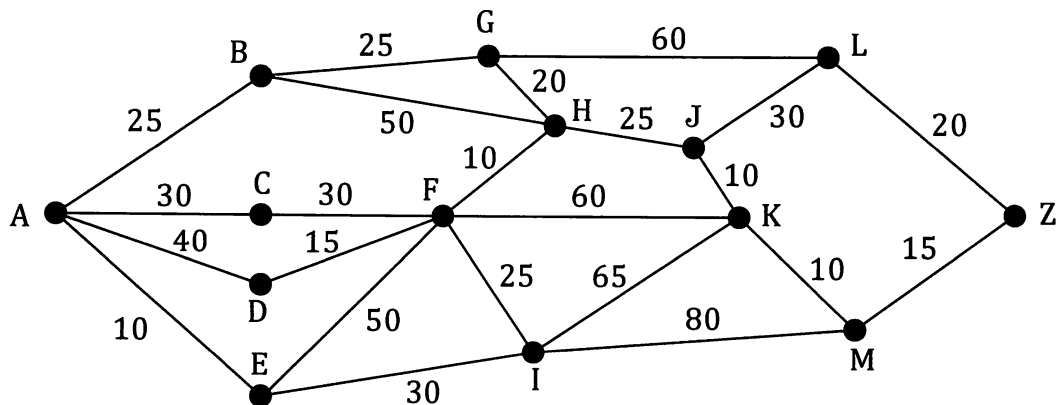
5.



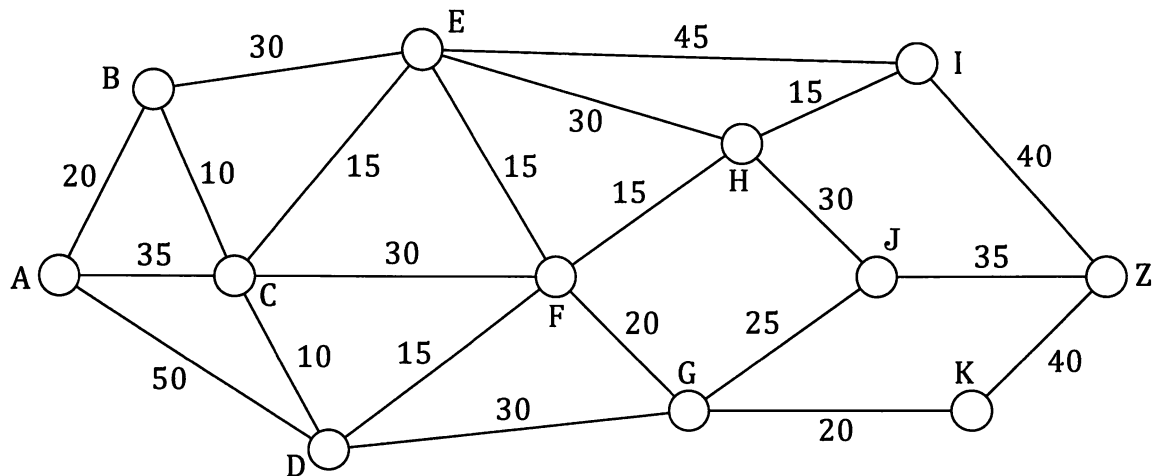
6.



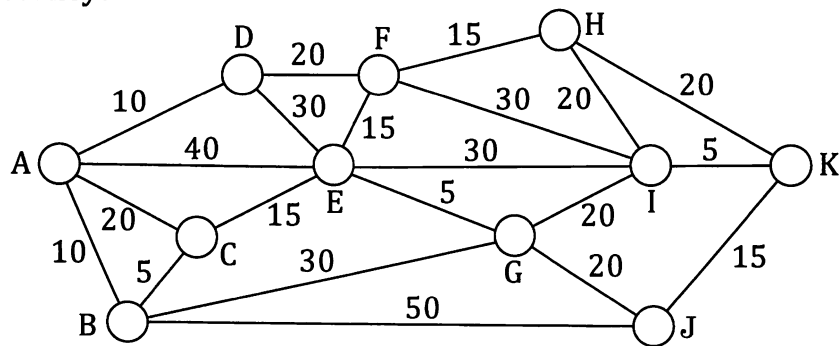
7.



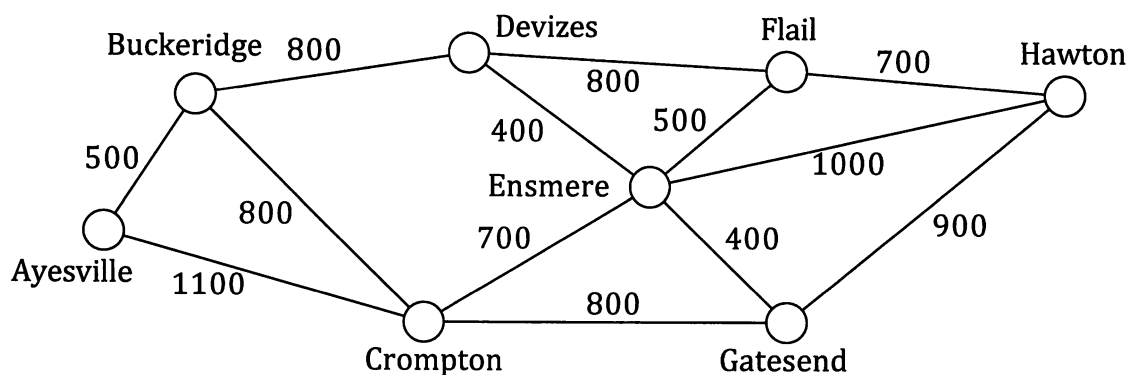
8.



9. The network shown below shows the time, in minutes, taken to travel along various sections of road. According to these times what is the quickest time that the journey from A to K could be completed in and what route would achieve this quickest journey?



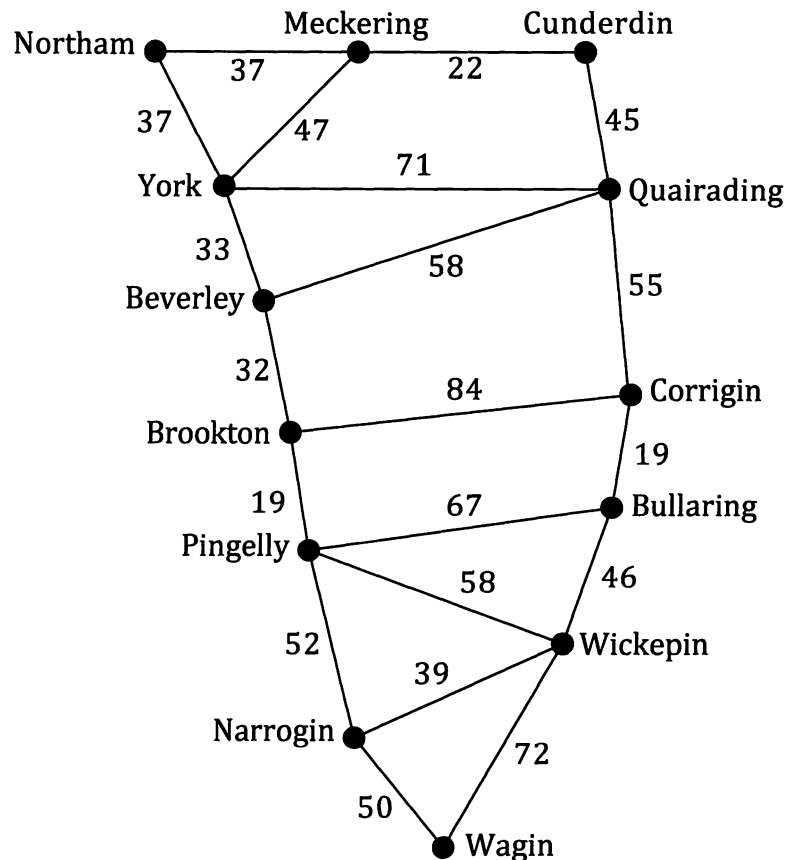
10. A decision is made to upgrade some of the roads between the villages of Ayesville and Hawton so that there is one completely upgraded route that motorists can take from Ayesville, all the way to Hawton. The diagram below shows the existing road sections and the numbers indicate the cost, in \$10000's, of upgrading each section. What route from Ayesville to Hawton should be chosen for upgrading if the cost of the upgrade is to be kept to a minimum? What will this minimum cost be?



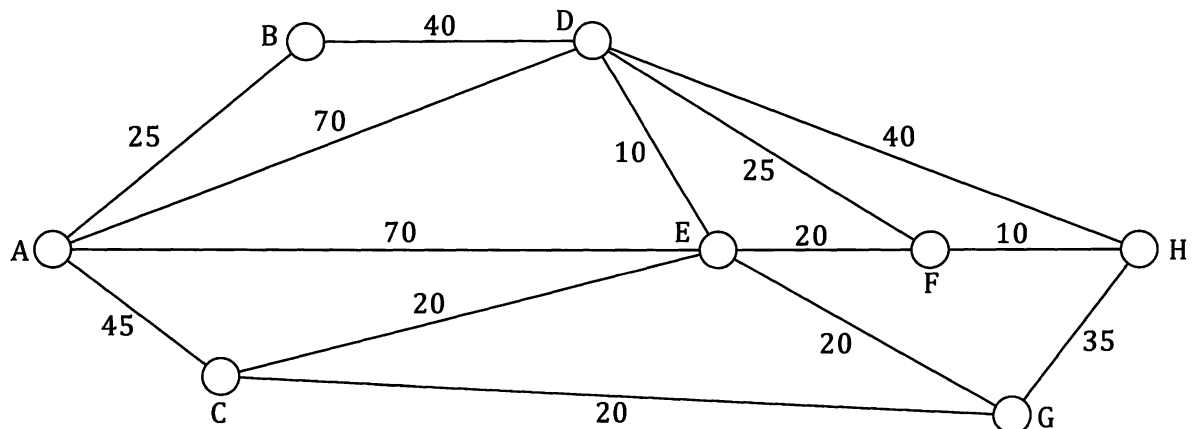
11. The network shown on the right shows the road distances, in kilometres, between a number of places in Western Australia.

What is the shortest route, and how far is it, from

- (a) Meckering to Wickiepin,
to
Wickepin,
- (b) Meckering to Wagin?

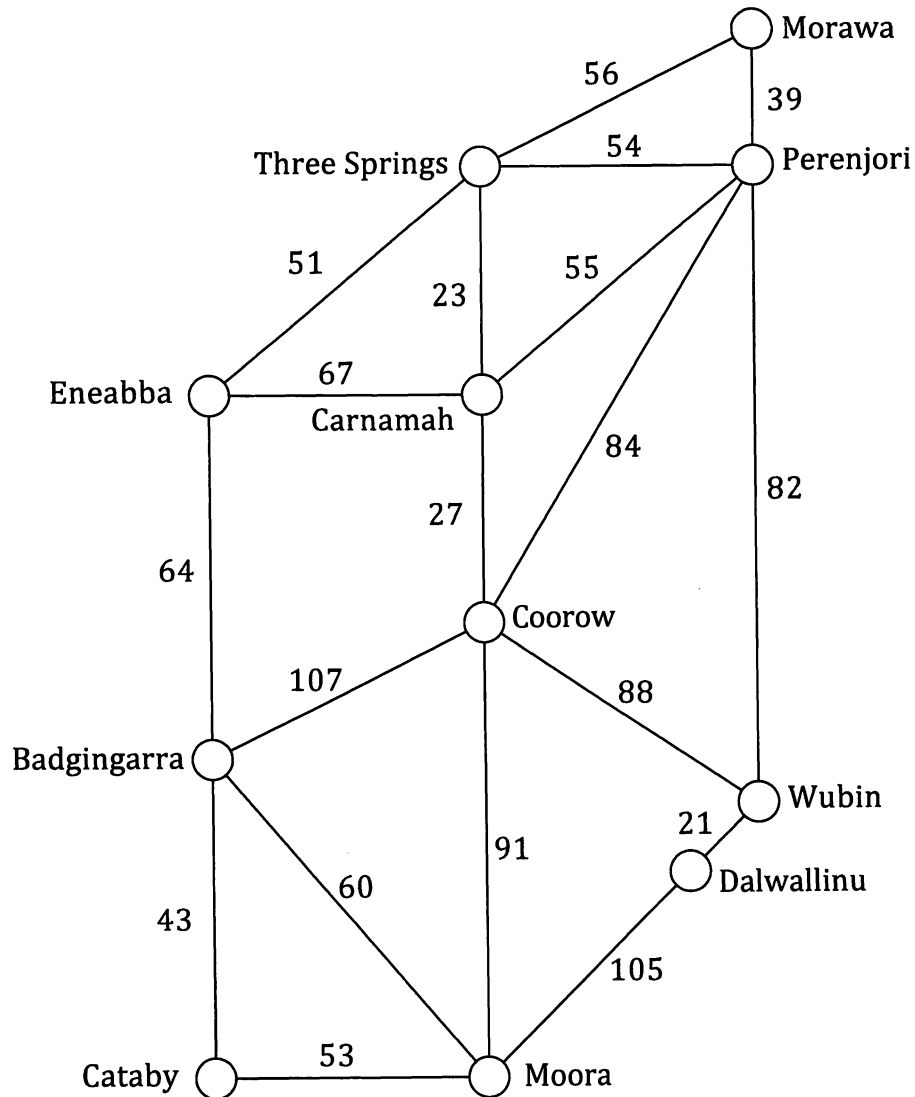


12. The network shown below represents the roads, shown as edges, linking a number of towns, shown as vertices. The numbers on each edge indicates the time, in minutes, required to travel that stretch of road.

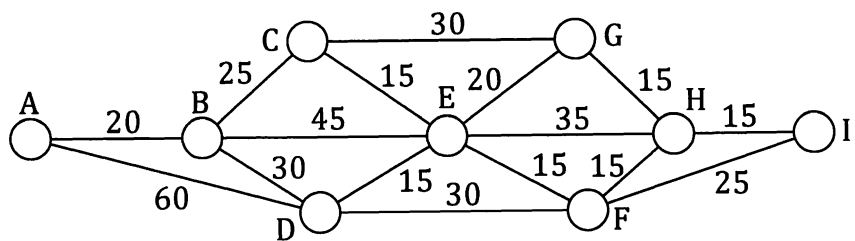


Determine the time and the route of the quickest journey possible from town A to town H, according to the given times.

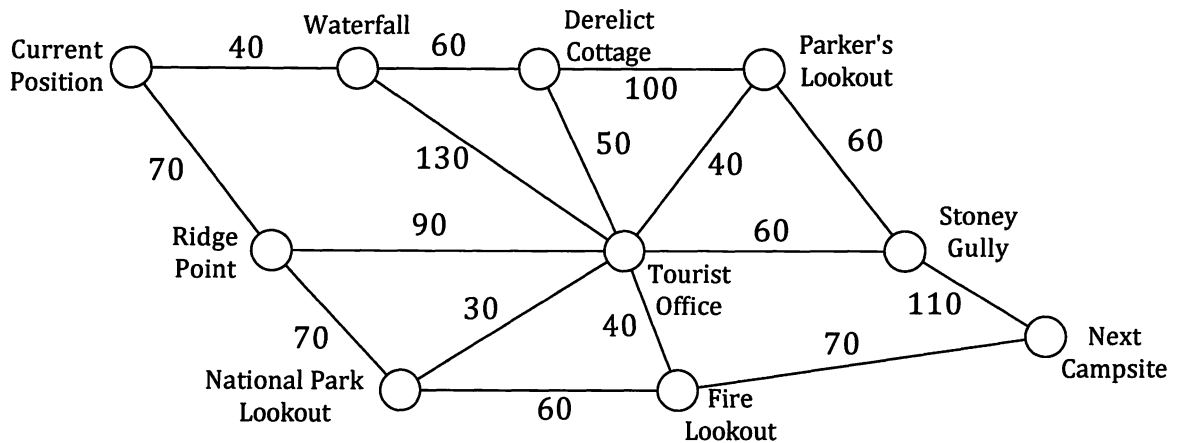
13. The network shown below, not drawn to scale, shows the road distances, in kilometres, between a number of locations. What is the shortest route from Cataby to Morawa along the roads shown, and how far is this route?



14. Find the shortest distance from A to I for the network on the right. State the route that gives this shortest distance.

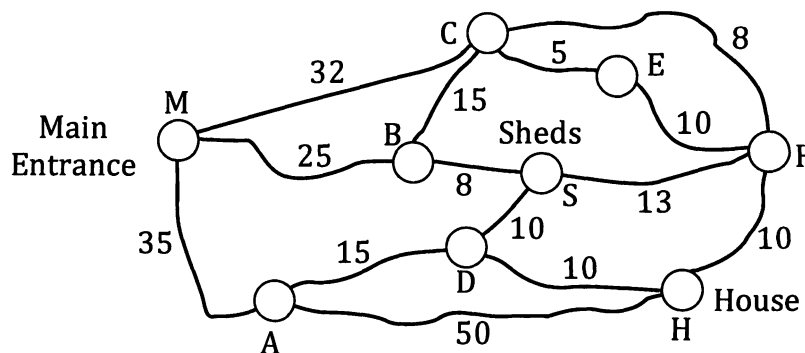


15. A hiker is planning her next day's journey. The network below shows the various tracks she can use to travel from her current position to the next campsite. Having an idea of the likely terrain and the condition of the tracks she estimates the time each section would take her, in minutes. According to these estimates (shown on the diagram) what route should she take in order to complete the journey as quickly as possible and, allowing an additional 30 minutes for a lunch break, how long should she expect the journey to take?



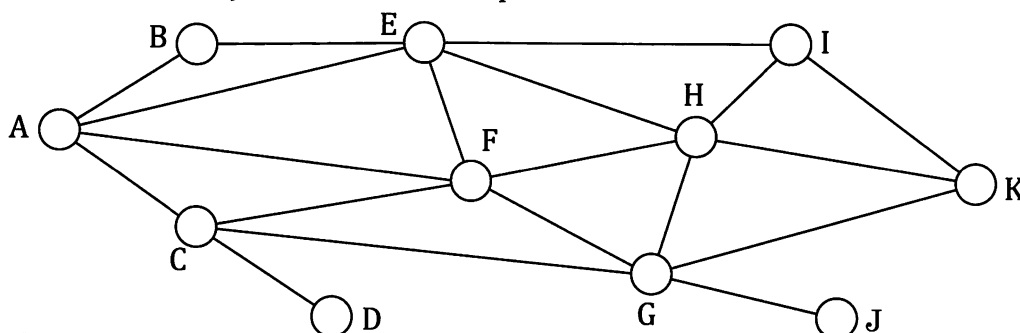
Having heard of the wonderful view at Parker's Lookout the hiker decides that she must include this in her route. What should her journey be now if it is to include Parker's Lookout and be completed as quickly as possible?

16. The diagram below shows the tracks running across a farming property. The owners of the property wish to have one route, from the main entrance to the house, upgraded. They ask a company to determine the cost of upgrading each track. Due to the variable quality of the existing tracks and the variable terrain that these tracks cross, the cost is not simply proportional to the length of the track. The numbers in the diagram indicate the cost of upgrading each track (in \$1000). What route should they choose to minimise the cost and what would this minimum cost be?



The owners then decide that the upgraded route must pass through junction S, where the farm sheds are situated. Does this requirement change the minimum cost and, if it does, determine the new route that incorporates the new requirement and keeps costs to a minimum? What would the minimum cost be now?

17. (a) Planners decide that a new road needs to be made through an area that is currently forestry. The planners decide that the best way to proceed is to upgrade some of the existing dirt tracks through the area to make one good route. The existing tracks are shown on the sketch of the area shown below (not drawn to scale) and the road is required from A to K.

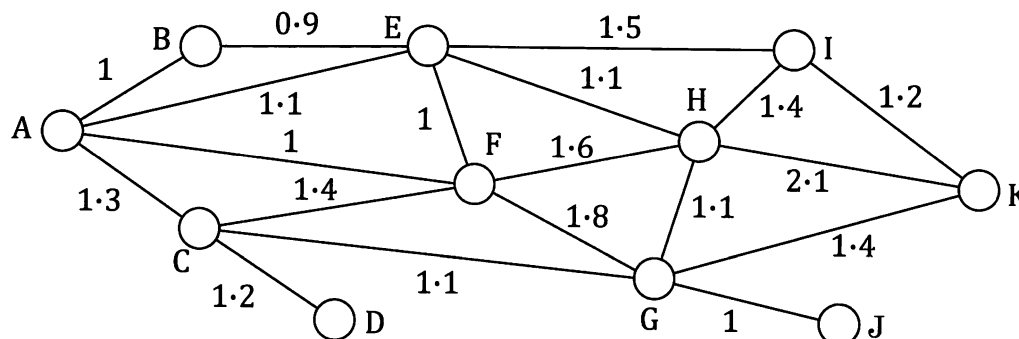


The length of each track, in metres, is given in the table below.

	A	B	C	D	E	F	G	H	I	J	K
A	-	210	320	-	600	510	-	-	-	-	-
B	210	-	-	-	420	-	-	-	-	-	-
C	320	-	-	140	-	280	460	-	-	-	-
D	-	-	140	-	-	-	-	-	-	-	-
E	600	420	-	-	-	180	-	210	350	-	-
F	510	-	280	-	180	-	240	200	-	-	-
G	-	-	460	-	-	240	-	240	-	350	440
H	-	-	-	-	210	200	240	-	180	-	500
I	-	-	-	-	350	-	-	180	-	-	350
J	-	-	-	-	-	-	350	-	-	-	-
K	-	-	-	-	-	-	440	500	350	-	-

What route from A to K should be chosen for upgrading to involve the minimum total length of track?

- (b) Further investigation reveals that some of the existing tracks will cost more per metre to upgrade than others. Taking the track AF as the standard, the diagram below gives the "cost per metre multiplication factor" for each track. For example to upgrade each metre of the track AC costs 1.3 times as much as it costs to upgrade 1 metre of AF.



What route should be chosen for the upgrade in order to minimise cost?

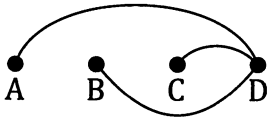
- (c) Before the upgrading commences a conservation group gets a ruling from the courts that the planned upgraded road must not pass through H. With this obeyed what route now gives the minimal cost?

Miscellaneous Exercise Six.

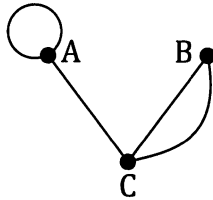
This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

Write the adjacency matrix for each of the following undirected graphs.

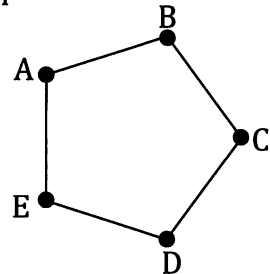
1.



2.

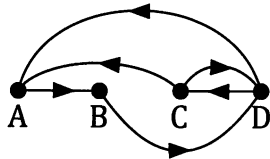


3.

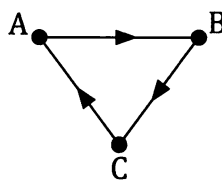


Write the adjacency matrix for each of the following digraphs.

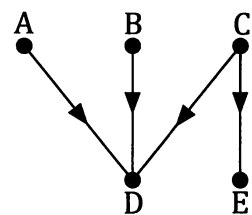
4.



5.

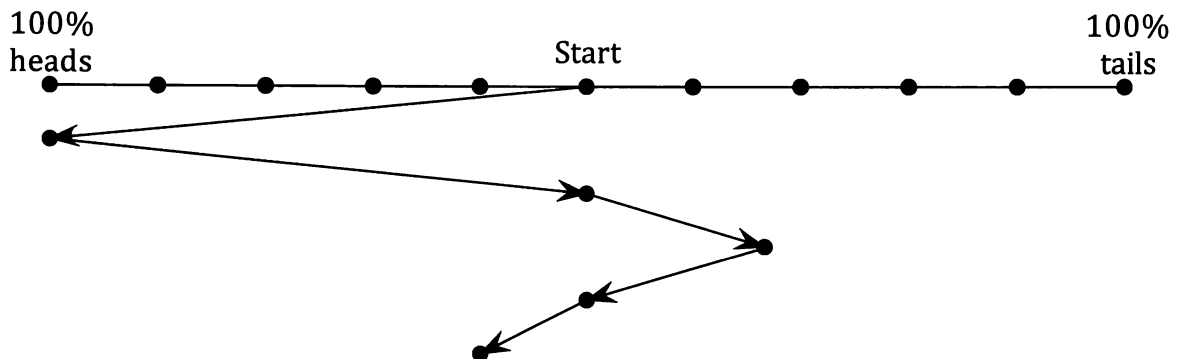


6.



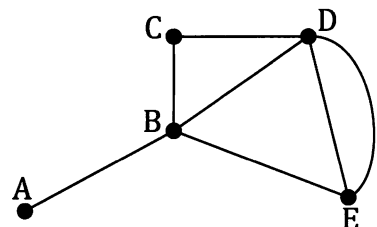
7. For question 6 above, simply by looking at the graph, determine what the square of the adjacency matrix will be.

8. The graph below was constructed for 5 flips of a fair coin.

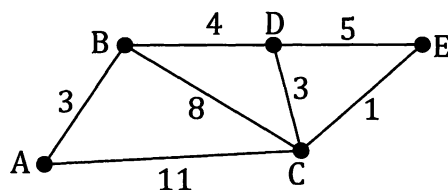


- What was the sequence of heads and tails for these five flips?
- Flip a coin 20 times and construct a similar diagram to that shown above for your results.

- Explain why it is correct to classify the graph on the right as being semi-Eulerian.
 - What edge needs to be added to the graph to make it Eulerian? Justify your answer.

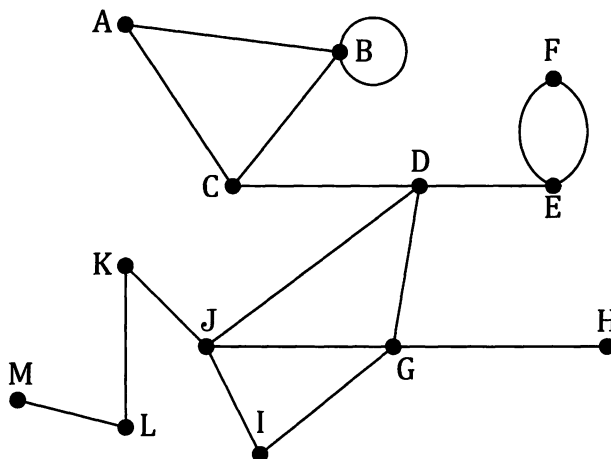


10. (a) For the weighted network shown on the right list all possible paths that start at A and finish at E, and state the total weight of each path. (Remember a *path* from A to E will have no repeat visits to vertices and no repeated travel along an edge.)

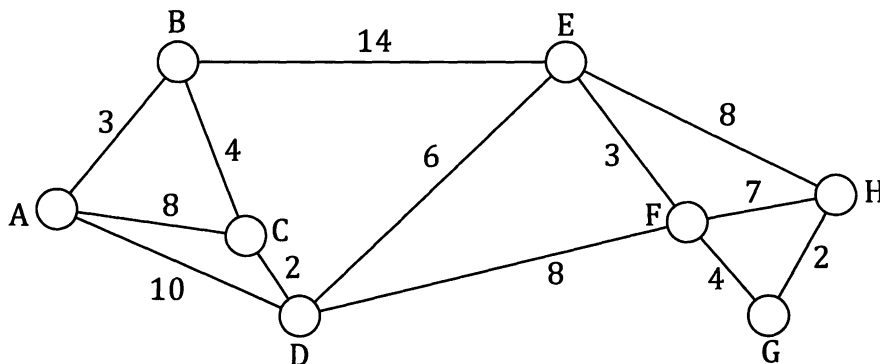


- (b) Which path from A to E has the least total weight?
 (c) Check that using the shortest path algorithm of this chapter gives the same answer as that obtained in part (b).

11. Which of the edges in the graph shown below are bridges?



12. Find the length (i.e. total weight) and route of the shortest path from A to H in the following weighted graph.



13. In each of the following sets of paired values what percentage of the variation in B can be accounted for, or explained by, the changes in A?

(a)

A	17	23	41	65	68	94	101	110	131	131	142	155
B	105	108	90	94	63	70	32	54	63	48	31	25

(b)

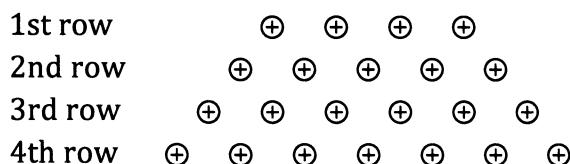
A	0.8	1.1	1.9	2.3	3.5	4.7	4.8	6.2	7.1	8.7	9.5	7.2
B	4.2	7.3	13.8	5.6	13.6	4.8	17.3	9.3	14.8	13.2	18.2	6.1

14. The number of stitches in each row of a particular crochet pattern are as shown on the right.

With T_1 the number of stitches in the first row, T_2 the number in the second row etc, express the sequence

$$T_1, T_2, T_3, T_4, T_5, \dots$$

using recursive notation (i.e. state T_1 and the recursive rule).



15. Determine by how much the eleventh term exceeds the tenth term in each of the following recursively defined sequences.

(a) $T_1 = 15, \quad T_{n+1} = T_n + 7.$

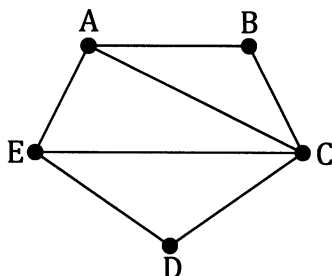
(b) $T_1 = 534, \quad T_{n+1} = T_n + 17.$

(c) $T_1 = 5, \quad T_{n+1} = 2 \times T_n.$

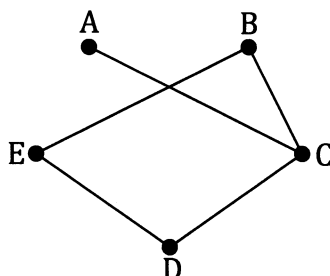
(d) $T_1 = 3, \quad T_{n+1} = 2 \times T_n + 4.$

16. Which of the following could be drawn as more obviously bipartite graphs?

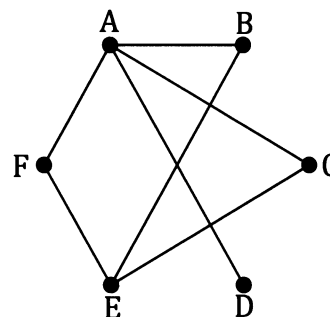
(a)



(b)



(c)



17. Fourteen students sat a mid-year exam and an end of year exam in a particular course. The results they gained were as follows:

Initials	KA	CB	JE	TE	TH	AJ	TK	RL	KM	NN	TP	DS	JS	RS	Out of
Mid Yr (M)	52	41	52	76	84	93	44	63	81	48	83	74	63	69	100
End Yr (E)	84	61	73	72	95	107	66	74	96	59	102	78	82	88	120

A fifteenth student missed the mid year exam but scored 65 out of 120 in the end of year exam.

- (a) Determine the correlation coefficient for the 14 pairs of results given above.

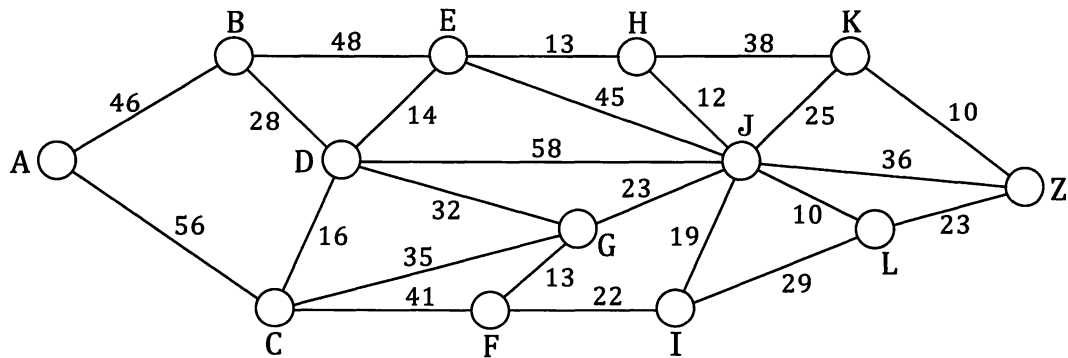
- (b) Determine the equation of the least squares regression line,

$$M = aE + b,$$

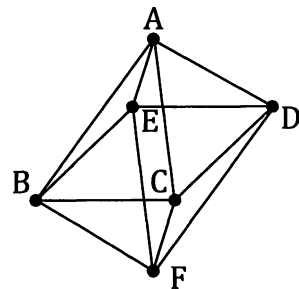
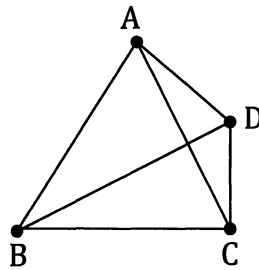
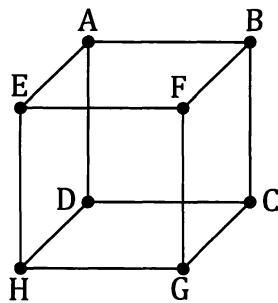
for predicting mid year marks based on end of year marks.

- (c) Use your regression line from (b) to estimate a mid year exam mark for the fifteenth student.

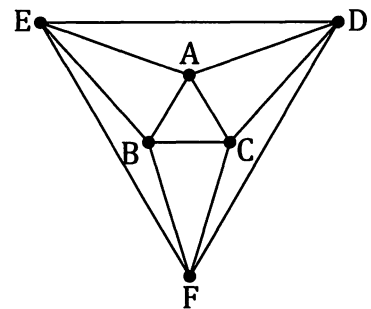
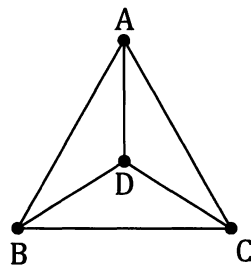
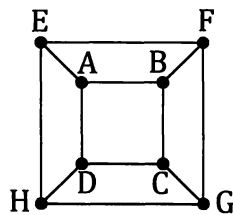
18. The diagram below shows the existing paths, tracks and roads through part of a forest. Some of these will be upgraded to produce one good quality road from point A to point Z. The numbers on the edges in the diagram show the cost, in thousands of dollars, of upgrading each path, track or road to the standard required. Determine the cost and route of the cheapest upgrade that will give one good quality road from A to Z.



19. The graphs below show a cube, a tetrahedron and an octahedron.



More obviously planar versions of these graphs are shown below.

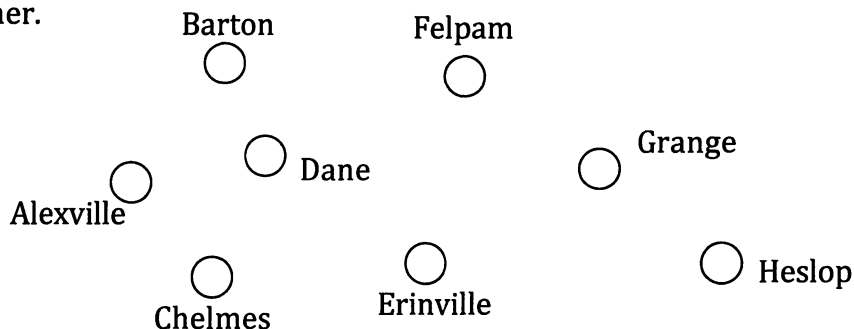


- Of these three planar graphs which are
- (a) connected graphs ,
 - (b) complete graphs ,
 - (c) Eulerian graphs ,
 - (b) Hamiltonian graphs?

20. The table below shows the times in minutes that it takes to travel between eight towns along the direct road links that exist between any two of the towns.

	Alexville	Barton	Chelmes	Dane	Erinville	Felpam	Grange	Heslop
Alexville	-	20	25	45	-	-	-	-
Barton	20	-	-	20	-	60	-	-
Chelmes	25	-	-	30	40	-	-	-
Dane	45	20	30	-	45	35	40	-
Erinville	-	-	40	45	-	-	20	50
Felpam	-	60	-	35	-	-	10	-
Grange	-	-	-	40	20	10	-	20
Heslop	-	-	-	-	50	-	20	-

The "map" below gives a rough idea of where the towns are situated with respect to each other.



Determine the shortest time that the journey from Alexville to Heslop can be completed in and state the towns you would pass through if you made the journey along the route that gives this shortest time.

21. The 437 students who completed a particular business management course came from two different colleges, Stack and Bosun.

Every one of the 437 students received a grade of either A, B, C, D or E.

241 of the students were from Stack and 77 of these received an A grade.

Altogether 145 B grades were awarded which was 44 more than the total number of A grades awarded.

The number of Stack college students receiving a B grade was 16 more than the number from Stack receiving an A.

Of the 158 students receiving a C grade, 60 were from Stack.

9 of the Stack students received a D.

The number of Bosun students receiving an E was 4 times the number of Stack students receiving an E.

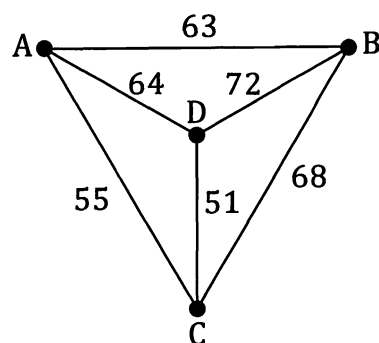
Copy and complete the following two way table showing the number of students from each college achieving each grade.

	A	B	C	D	E	Totals
Stack College						
Bosun College						
Totals						437

Investigate whether there seems to be an association between the grade achieved and which of the two colleges the student comes from.

The data given in this question is fictional and does not relate to any colleges in real life.

22. The weighted graph shown on the right gives the estimated times, in minutes, required to travel on foot between the various pairs of locations that can be chosen from four locations, A, B, C and D situated in a challenging, rugged, outback environment. One possible Hamiltonian circuit starting and finishing at A is ACBDA with a total estimated time of 259 minutes.



- (a) For the estimated times given is this the quickest Hamiltonian circuit for this graph? Investigate.
- (b) In this situation, but not necessarily with the times as stated, if we must start at A, visit each of B, C and D and then end up back at A, will it always be a Hamiltonian circuit that gives us the shortest time or will it depend on the estimated times?
23. A person opens an account with a deposit of \$400 000. The account pays interest of 6.7% per annum with the interest added at the end of each twelve month period. At the same time as interest is added the person withdraws \$55 000 per year for as many years as possible and, at the end of the first year that such a withdrawal is not possible, the person plans to withdraw the remaining amount and close the account. For how many years can the \$55 000 be withdrawn and what will the final withdrawal amount be?

Extension.

24. A farmer is developing his herd of a particular species of animal that is not commonly farmed, with the intention of selling them for meat and other products. The farmer currently has 500 animals in the herd and, allowing for normal birth and death rates, anticipates an annual growth of 10% (compound) before any are culled for market.

The farmer wishes to consider the following ideas.

- ① Cull a regular number at the end of each year, say 40.
- ② Cull an increasing number each year, say

40	at the end of the 1 st ,
80	at the end of year 2,
120	at the end of year 3,
160	at the end of year 4,
	etc.
- ③ Cull an increasing number each year say

40	at the end of year 1,
40×1.2	at the end of year 2,
40×1.2^2	at the end of year 3,
40×1.2^3	at the end of year 4,
	etc.

Explore each of these possibilities in terms of their effects on the long term population. Consider varying the "40" that appears in each one. Consider varying the rate of increase in the third possibility. Consider

Write a report for the farmer.

Answers.

Exercise 1A. Page 21.

1. (a) 115 people responded to the survey.
 (b) 28 of the respondents chose beef as their most preferred main course at a restaurant.
 (c) 32% (nearest percent) of those choosing beef as their most preferred course at a restaurant were aged 31 to 43.
 (d) 26% (nearest percent) of those aged 31 to 43 chose beef as their preferred main course at a restaurant.
2. (a) The frequency table involves 376 people.
 (b) The frequency table involves 162 males.
 (c) 15% (nearest percent) of the males involved in the survey were divorced.
 (d) 44% (nearest percent) of the divorced people involved in the survey were males.
 (e) According to the survey results the percentage of males who are single (30%) is greater than the percentage of females who are single (28%).

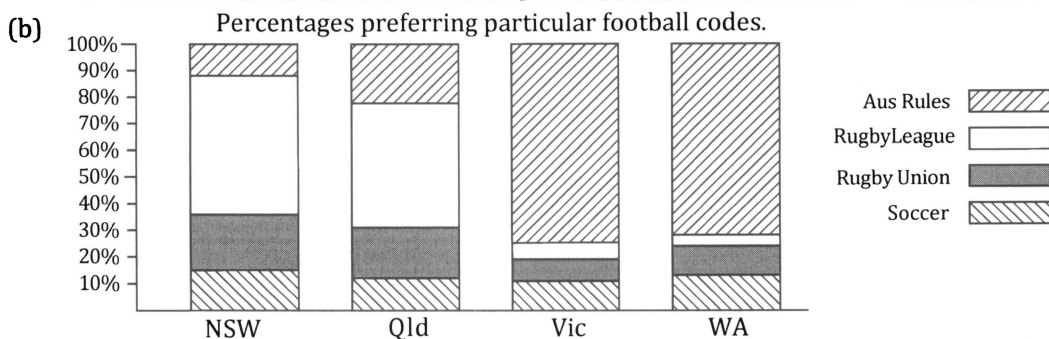
3. (a)		A	B	C	D	Totals
	E	11%	13%	17%	37%	78%
	F	6%	7%	7%	2%	22%
	Totals	17%	20%	24%	39%	100%

(b)		A	B	C	D	Totals
	E	14%	16%	22%	48%	100%
	F	26%	33%	32%	9%	100%

(c)		A	B	C	D
	E	65%	64%	71%	95%
	F	35%	36%	29%	5%
	Totals	100%	100%	100%	100%

For question 4, 5 and 6 the comments given here are by way of example, they are unlikely to be the only correct comments that could be made. However for each question your comments should include mention of whether the given data suggests association between the variables, as each question specifically asked for such comment.

4. (a)		NSW	Queensland	Victoria	WA
	Australian Rules	12%	22%	75%	72%
	Rugby League	52%	47%	6%	4%
	Rugby Union	21%	19%	8%	11%
	Soccer	15%	12%	11%	13%
	Totals	100%	100%	100%	100%



- (c) In the graph, as we move across the state columns, whilst some similarities exist there are also some very clear differences in the proportions preferring each sport, thus suggesting there is an association between the two variables, preferred football code and state lived in. New South Wales and Queensland show similar distributions with each of these states having Rugby League as the most preferred code. 52% of NSW respondents choosing this and 47% of the Queensland respondents.

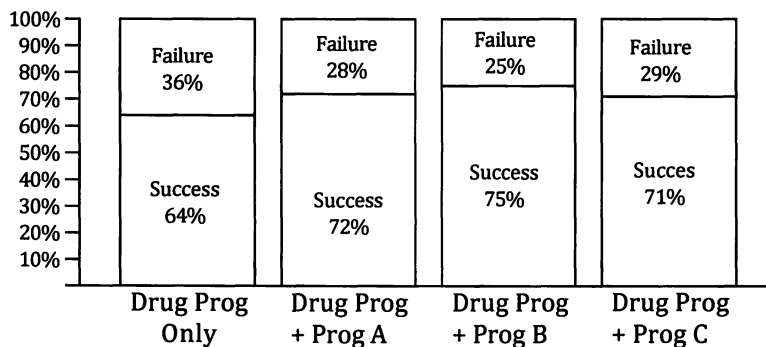
Victoria and West Australia were also similar to each other, both having Australian rules as the preferred option and both putting this well ahead of the other options. (75% of Victorian respondents and 72% of West Australian respondents).

However the "NSW-Qld" distribution and the "Vic-WA" distribution show considerable differences between each other. Whilst NSW and QLD had Rugby League as easily the most preferred code, Victoria and West Australia placed Rugby League as least preferred and had Australian rules as the most preferred by a considerable margin.

5. (a)

	Success	Failure	Totals
Drug program only	64%	36%	100%
Drug program + program A	72%	28%	100%
Drug program + program B	75%	25%	100%
Drug program + program C	71%	29%	100%

(b)



- (c) There seems to be a reasonably consistent increase in the success rate of the program if a sufferer is on one of the programs, almost irrespective of which program, A, B or C, they are on. Hence there does seem to be an association between the success rate and whether or not a sufferer is on any one of the additional programs or not.

Sufferers who take part in one of the three additional programs, A, B or C have a 71% to 75% success rate compared to 64% proportion of success without one of the additional programs. According to the figures given in the table Program B has the greatest increase followed by A then C but the variation between the A, B and C results is not great.

(Note: We cannot necessarily assume that the additional programs are *causing* the increased success. It could be that people who opt for an additional program are the type more likely to succeed anyway. This *cause* aspect will be considered further in chapter 2.)

6. (a)

	30 and under	31 to 40	41 to 50	51 to 60	61 & over
Male	48%	53%	52%	68%	71%
Female	52%	47%	48%	32%	29%
Totals	100%	100%	100%	100%	100%

- (b) The proportions of male and female within each age category do not alter much as we move across the first three age categories and in each of these there is an approximately 50 – 50 split between males and females. Hence for these younger age categories there does not seem to be an association between age and gender balance in this workplace. However the proportions change quite dramatically in the final two categories where our previously roughly 50 – 50 split becomes male dominated with more than 2 males to each female in each of these last two categories. Thus there does seem to be an association between the age categories of under 50 and over 50 and gender balance in this work place.

We are told that the organisation has in recent years introduced a workplace gender equality policy so perhaps this, maybe coupled with a change in society views, has led to the more equal gender balance in the younger, and therefore perhaps more recently appointed, employees.

- 7, 8 and 9. For question 7, 8 and 9 answers are not given here. Compare your answers with those of others in your class and with your teacher. Comment on the responses of others to these questions and allow them to comment on your responses.

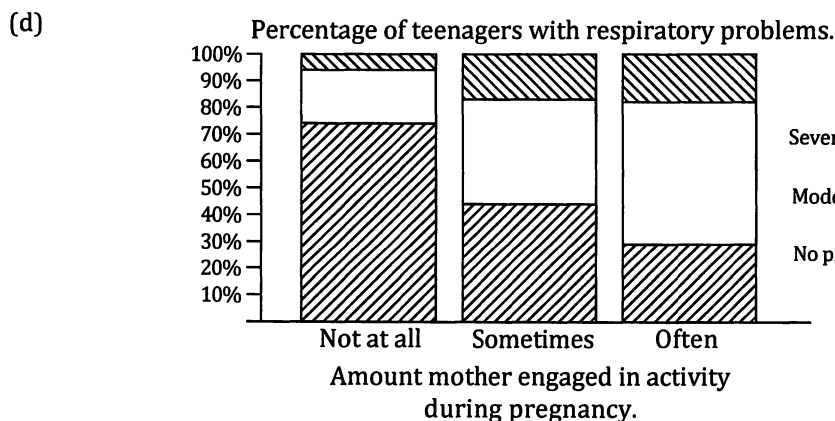
Exercise 1B. Page 27.

1. Explanatory variable: Gender.
Response variable: Preferred film type.
2. Explanatory variable: Gender of parent.
Response variable: Preferred activity of daughter.
3. Explanatory variable: Age of mother when person was born.
Response variable: Likelihood of person developing respiratory problems.
4. Explanatory variable: Left or right handedness of mother.
Response variable: Left or right handedness of child.
5. Explanatory variable: State a person lives in.
Response variable: Preferred football code.
6. Explanatory variable: Amount of study.
Response variable: Grade in exam.
7. Explanatory variable: Population of country.
Response variable: Number of babies born in country per year.
8. Explanatory variable: How well a person sleeps at night.
Response variable: The person's grumpiness the next morning.
9. Explanatory variable: Having, or not having, the mental disorder.
Response variable: Hearing voices.
10. Explanatory variable: Gender.
Response variable: Frequency of brushing teeth.
11. Explanatory variable: Smoking classification.
Response variable: Respiratory function.
12. Explanatory variable: Age of women's mother when had first child.
Response variable: Age of women when she has her first child.
13. Explanatory variable: The time the heat has been applied for.
Response variable: The temperature of the milk.
14. Explanatory variable: Lack of sleep.
Response variable: Level of tiredness.
15. Explanatory variable: Stress levels.
Response variable: Heart disease.
16. Explanatory variable: How much alcohol the driver has consumed recently.
Response variable: How dangerous the driver is to other road users.

17. (a) The investigation is likely to indicate whether respiratory problems in teenage years could in any way be explained by the amount that the mother of the teenager engaged in a particular activity during pregnancy. The amount the mother engaged in the activity during pregnancy is the explanatory variable and the level of respiratory problems experienced by the child in their teenage years is the response variable.
- (b) The explanatory variable (level mother was engaged in the activity during pregnancy) have categories "not at all", "sometimes" and "often". These are displayed using different rows.

(c)

		Level of respiratory problems in teenage years			Totals
		None	Moderate	Severe	
Engaged in activity during pregnancy	Not at all	74%	20%	6%	100%
	Sometimes	44%	39%	17%	100%
	Often	29%	53%	18%	100%



- (e) As we move across the explanatory variable categories "not at all", "sometimes" and "often" the big differences between columns in the percentages in the response variable categories indicate that there is an association between the two variables.

Further comments: (Given by way of example.)

The proportion of teenagers with no respiratory problems in their teenage years decreases dramatically as the engagement of the mothers in the activity during pregnancy increases. The proportion goes from 74%, for those teenagers with mothers who did not engage in the activity at all during pregnancy, to just 44% for those teenagers with mothers who engaged in the activity sometimes during pregnancy, to 29% for those teenagers with mothers who often engaged in the activity during pregnancy.

The percentage of teenagers with "moderate" respiratory problems increases as the engagement of the mother in the activity during pregnancy increases.

For mothers who engage in the activity during pregnancy, the likelihood of their child having severe respiratory problems in the teenage years is roughly treble that of a child whose mother does not engage in the activity (from 6% to about 18%). However, the figures suggest that this higher figure is little changed by whether the mother's engagement in the activity was sometimes or often.

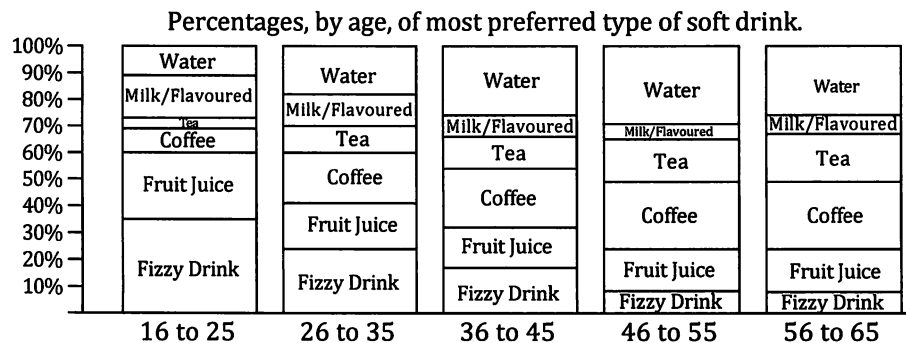
(Note: We cannot necessarily assume that it is the mother's engagement in the activity that is *causing* the respiratory problems of the child. It could be that the genetic make up of mother and child is responsible for both the mother's desire to take part in the activity and the child's respiratory problems. This *cause* aspect will be considered further in chapter 2.)

18. (a) We are investigating whether an individual's preferred soft drink can be explained by the age of the individual. Age is the explanatory variable and preferred choice of soft drink the response variable.

(b)

	Number of people nominating the drink as most preferred soft drink.						Totals
	Fizzy Drink	Fruit Juice	Coffee	Tea	Milk/ Flavoured Milk	Water	
16 to 25	35%	25%	9%	4%	16%	11%	100%
26 to 35	24%	17%	19%	10%	12%	18%	100%
36 to 45	17%	15%	22%	12%	8%	26%	100%
46 to 55	9%	15%	25%	16%	6%	29%	100%
56 to 65	8%	16%	25%	18%	7%	26%	100%

- (c) Whilst a graph is not specifically requested in the question it is shown below to aid interpretation.



As we move across the age categories changes in the proportions of most favoured drink are very evident. For example the fizzy drink category declines from being the drink most favoured by 35% of people in the 16 to 25 age group to just 8% of people in the 56 to 65. This change in the percentages as we move across the age brackets indicates that there is an association between age and preferred choice of soft drink.

Further comments: (Given by way of example.)

As mentioned above, the percentage of people having fizzy drinks as their most preferred choice of soft drinks declines considerably with age from 35% (16 to 25 yr age group), to 24% (26 to 35 yr age group) to 17% (36 to 45 yr age group) to 9% (46 to 55 yr age group) to 8% (56 to 65 age group).

As the age group considered increases:

- the percentage of people choosing fizzy drinks as most preferred soft drink decreases,
- the percentage of people choosing tea and coffee as most preferred soft drink increases,
- the percentage of people choosing water as most preferred soft drink increase to the 36 to 45 age group, after which it maintains a reasonably steady level.

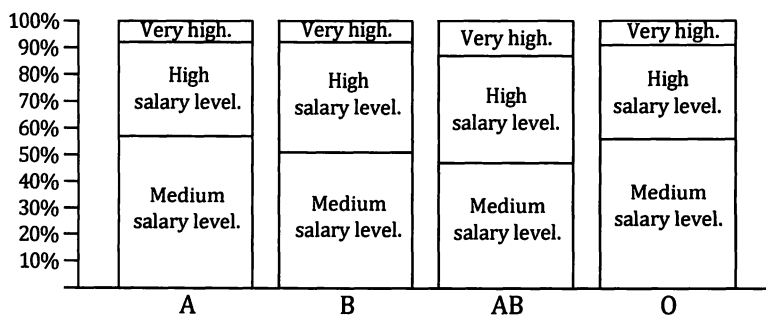
With the exception of the fizzy drink decline in popularity, the last three age groups, 36 to 45, 46 to 55 and 56 to 65 have very similar percentage distributions. For these age groups water and coffee are almost equally the most preferred, at around the 25% popularity level, with water generally just ahead of coffee. Tea and Fruit Juice are next, each at around the 15% level. This is in contrast with the 16 to 25 yr age group where fizzy drink and fruit juice dominate whilst tea and coffee are least preferred.

19. (a) We would expect some variation in salary due to being in work part time or full time. Similarly we might expect some variation in salary due to age, with an older worker possibly being paid more for their experience and seniority. Limiting the survey to those in full time employment and those aged between 30 and 40 attempts to remove from the collected data, or at least substantially reduce, salary variations due to these factors.
(However, with the reasons for salary variation being considered then the age of the workers should be considered at some stage as that aspect is likely to be a factor.)
(Part time workers could perhaps be included if their salary details were appropriately adjusted to be comparable to a full time worker.)
- (b) We are investigating whether salary level can in any way be explained by blood group. Blood group is the explanatory variable and salary level the response variable. For the given table we need to calculate row percentages. (That will allow us to see if the salary level proportions change as we move across the blood groups.)

		SALARY			Totals
		Medium	High	Very High	
BLOOD GROUP	A	57%	35%	8%	100%
	B	51%	41%	7%	99%*
	AB	47%	40%	13%	100%
	O	56%	35%	9%	100%

* The total of 99%, rather than 100%, is simply due to rounding.

GRAPH (not essential but shown here to aid interpretation):



Comments: (Given by way of example.)

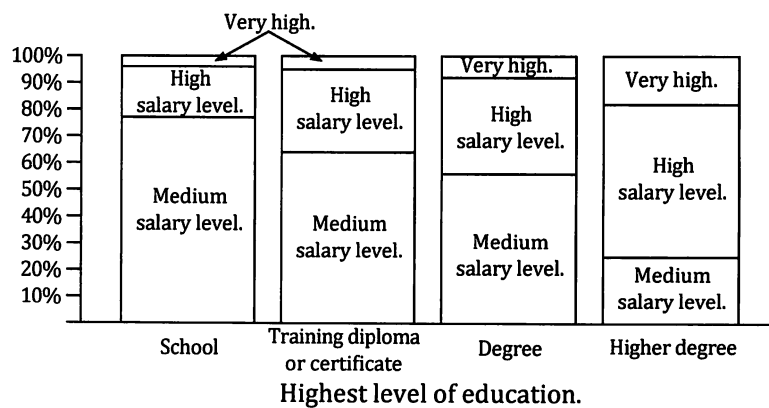
The percentages in the various salary levels change only a little as we move across the blood groups. The AB group proportions differ most from the others, especially at the Very High Salary category but the AB group involved a very small number of people compared to the other categories (15 compared to 41, 150 and 190). Just one of these 15 in a different salary level could significantly change the AB group percentages. If we look at the two biggest blood groups, A and O, the proportions in each salary category are almost identical.

The figures suggest that for people aged between 30 and 40 working full time in this particular field there is no association between the person's blood group and their salary level.

- (c) We are investigating whether salary level can in any way be explained by level of education. Level of education is the explanatory variable and salary level the response variable. For the given table we need to calculate column percentages. (That will allow us to see if the salary level proportions change as we move across the education categories.)

		HIGHEST LEVEL OF EDUCATION			
		School	Training Diploma or Certificate	Degree	Higher Degree
Salary	Medium	77%	64%	56%	25%
	High	19%	31%	36%	57%
	Very High	4%	5%	8%	18%
	Totals	100%	100%	100%	100%

GRAPH (not essential but shown here to aid interpretation):



COMMENTS (given by way of example)

As we move across the categories of highest level of education, from school through to higher degree:

- the percentage of people on medium level salary decreases (77%, 64%, 56%, 25%),
- the percentage of people on high level salary increases (19%, 31%, 36%, 57%),
- the percentage of people on very high salary increases (4%, 5%, 8%, 18%).

The changing proportions indicate a strong association exists between salary levels and highest level of education.

The data suggests that for people in full time employment and aged between 30 and 40, the higher a person's level of education completed, the greater the likelihood of that person being on one of the higher salary levels.

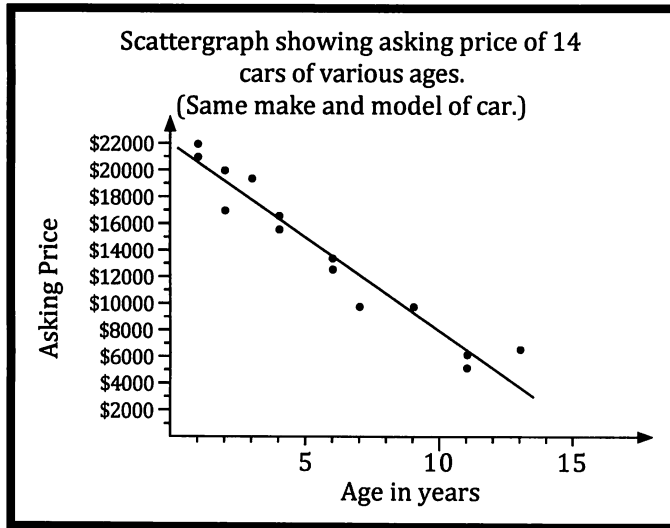
Whilst three quarters of those who had school as their highest level of education were on the medium salary level, only one quarter of those with a higher degree were on this salary level, the other three quarters being on high or very high salaries.

- 20 and 21.** For questions 20 and 21 answers are not given here. Instead compare your answers with those of others in your class. Read and comment on their report and allow them to read and comment on yours. Ask your teacher to read and comment on your report and on the comments others have made about your report.

Exercise 1C. Page 35.

Note: When judging the line of best fit "by eye" on a scattergraph there can sometimes be considerable variation between one person's judgement and another's. As you will see later in this unit there is a way in which the line of best fit can be calculated mathematically. It is this calculated line of best fit that is shown in these answers. Any predicted values you give should be based on *your* lines of best fit which should not be too far from the lines shown here.

1.



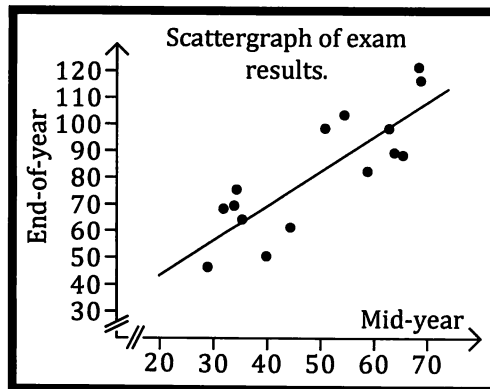
Asking price falls as age of vehicle increases.

Distribution of points suggests a linear model would be a suitable "fit".

Likely price for a five year old model:

Approximately \$14900.

2. (a) and (c)

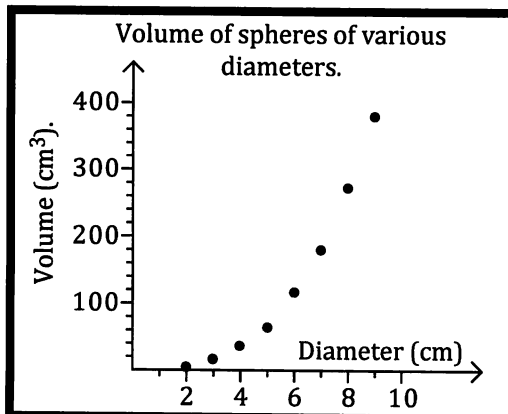


(b) Students with higher mid-year results tended to gain the higher end-of-year results and students with lower mid-year results tended to gain the lower end-of-year results.

A linear model would be a reasonable one to use for estimates (accepting of course that any estimates involving extrapolation would have to be viewed with particular caution).

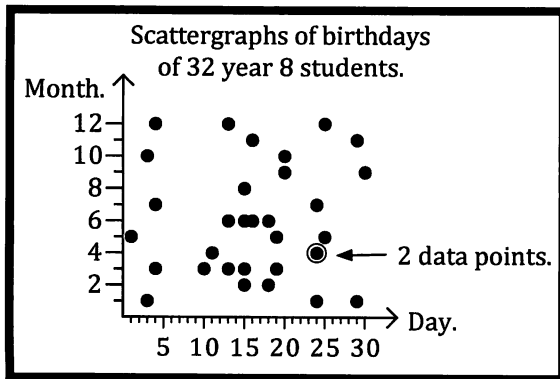
(d) Approximately 82

3. (a)



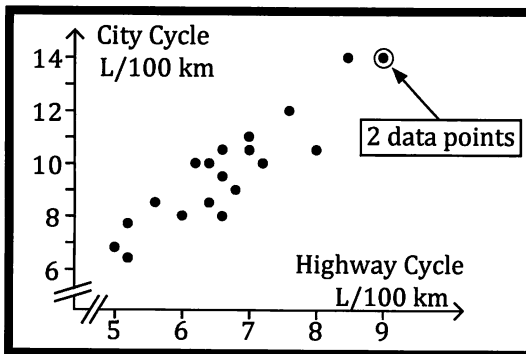
(b) Graph indicates that a relationship exists between diameter and volume but this relationship is not linear. Hence a linear model would not be a good choice for this data.

4. (a)



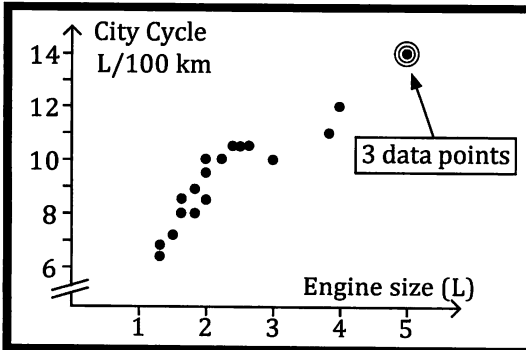
Graph indicates no relationship exists between day of birth and month of birth.

5. (a)



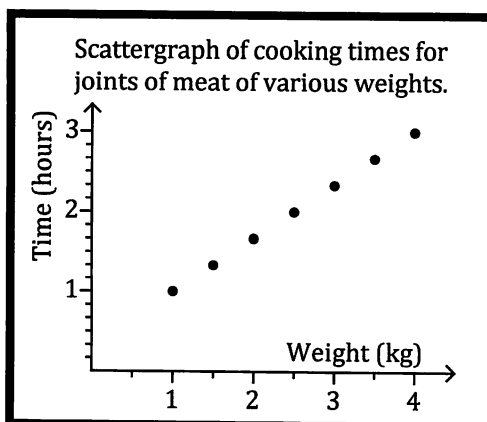
Cars that use more litres of fuel per 100 km of highway cycle also tend to use more fuel per 100 km of city cycle. The relationship between these two variables could be well represented by a linear model.

(b)



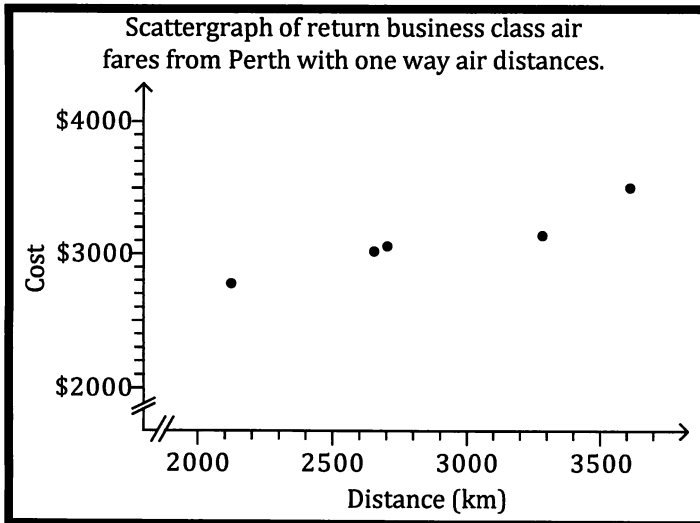
Cars with the bigger engine size tend to use more litres per 100 km on city driving than those with the smaller engines. Whilst the relationship between engine size and litres per 100 km could be quite well represented using a single linear model the graph perhaps better fits two straight lines, one for engine size up to about 2.8 L and then one for engine size from 3 litres upwards.

6.



Graph indicates that the cooking time is perfectly linearly related with the weight of the joint of meat with cooking time increasing 20 minutes for each 500 gram increase in weight.

7.

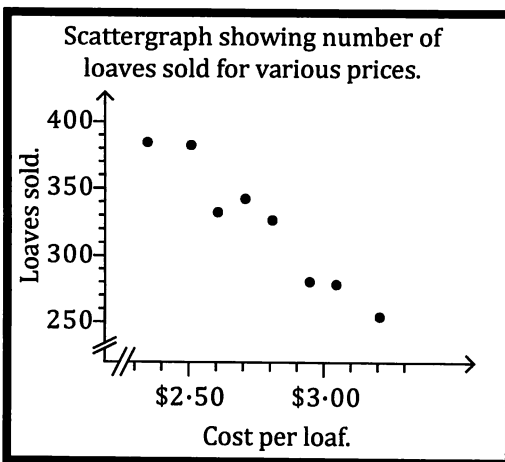


Only a few data points so difficult to have great confidence in any relationship suggested but the points we have do seem to indicate a steady increase in the cost as the distance increases. A linear model would fit the few data points well.

(b) Linear fit would suggest approx \$2900. This involves interpolation so could be reliable but would like more data points before being confident of linear model.

(c) Extrapolating to 15000 km using a linear model suggests a fare of approx \$8000 to \$9000 but we are then way beyond the known data points so could have very little confidence in this estimate.

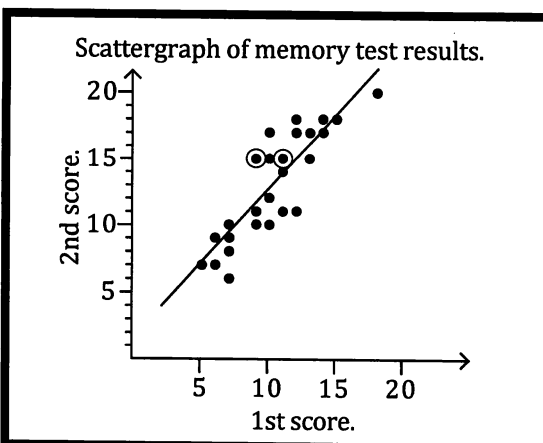
8.



The number of loaves sold steadily decreases as the cost per loaf increases. A linear model would fit the relationship well.

A linear model would suggest that for a price of \$2.90 the baker needs to bake approx 300 loaves.

9.



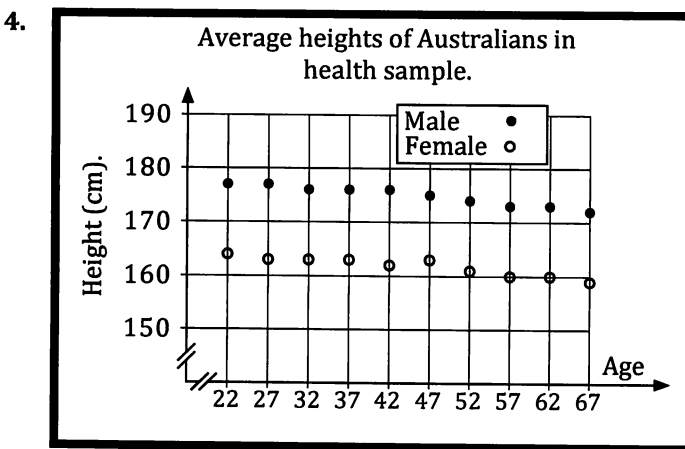
Those who scored the higher marks in the first test tended to be the ones who also scored the higher marks in the second test.

Predicted second score for a first score of 8 would be approximately 10 or 11.

Miscellaneous Exercise One. Page 39.

1. (a) 24.5 (b) 340 (c) 112 (d) 24 (e) 36
2. The 10 means that for each one kg of extra weight suspended from the spring its length increases by 10 cm.
The 25 means that the unstretched length of the spring (i.e. the length with no weight suspended from it) is 25 cm.

	Coordinates of y -axis intercept.	Gradient	Equation
A	(0, 2)	1	$y = x + 2$
B	(0, -1)	2	$y = 2x - 1$
C	(0, -3)	-2	$y = -2x - 3$
D	(0, 3)	-1	$y = -x + 3$
E	(0, 2)	0.5	$y = 0.5x + 2$
F	(0, -2)	0.5	$y = 0.5x - 2$
G	(0, 1)	-0.5	$y = -0.5x + 1$



The average height of the male Australians in the sample was consistently about 13 centimetres more than the average height of the Australian females of the same age.

The graph suggests that for both males and females the trend is for the younger adults in society to have a greater average height than the older folk.

5. To investigate whether the proportions of each type of crime change much as we move across the categories of alcohol consumption we want crime proportions within each alcohol consumption category. Hence we create column percentages.

	Usual alcohol consumption.			
	None	Light	Medium	Heavy
Assault	10%	8%	20%	32%
Fraud	24%	22%	14%	7%
Motor vehicle	52%	40%	39%	30%
Theft (not involving assault)	14%	30%	27%	32%
Totals	100%	100%	100%	101%*

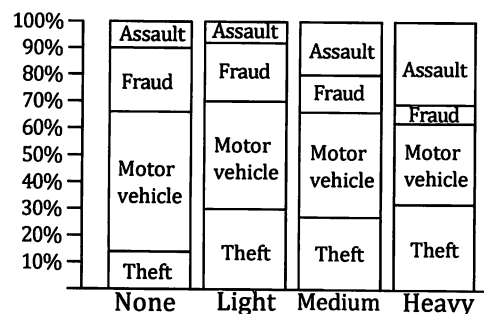
* Rounding anomaly.

As we move across the categories of alcohol consumption the proportions of each crime type change quite noticeably, thus suggesting there is an association between alcohol consumption and type of crime.

The proportion of crimes that were crimes of assault was much higher (32%) amongst heavy drinkers than for non or light drinkers (10% and 8%).

The proportion of crimes that were crimes of fraud was much higher amongst non & light drinkers (24% and 22%) than among heavy drinkers (7%).

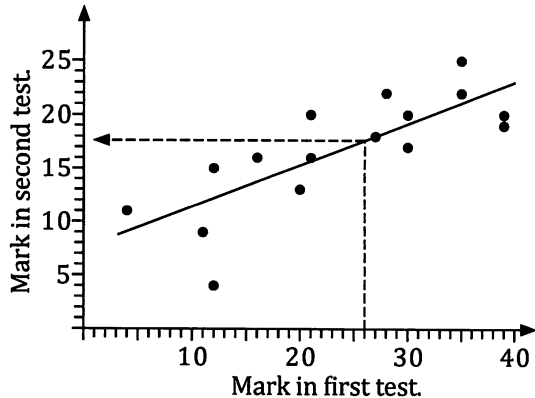
Crimes of theft had a smaller proportion amongst the non drinkers (14%) than in the other categories (~30%).



Exercise 2A. Page 44.

- The -1.5 suggests that on average each 1 year increase in the age of the child sees the time taken to complete the task decrease by 1.5 seconds.
The 41 suggests that a child aged 0 years would complete the task in 41 seconds. However, not only does the concept of "age zero" not make sense in this situation, it is also outside of the age range considered, i.e. 5 to 18.
- The 2 suggests that on average each one hour of revision sees an increase in the exam mark of 2%.
The 28 suggests that with no revision the mark gained would be 28%.
- Scattergraph A: $y = 0.271x + 48.9$ Scattergraph B: $y = 0.283x + 26.4$
Scattergraph C: $y = -0.490x + 49.0$ Scattergraph D: $y = 0.283x + 20.7$
Scattergraph E: $y = -0.152x + 48.6$

4.

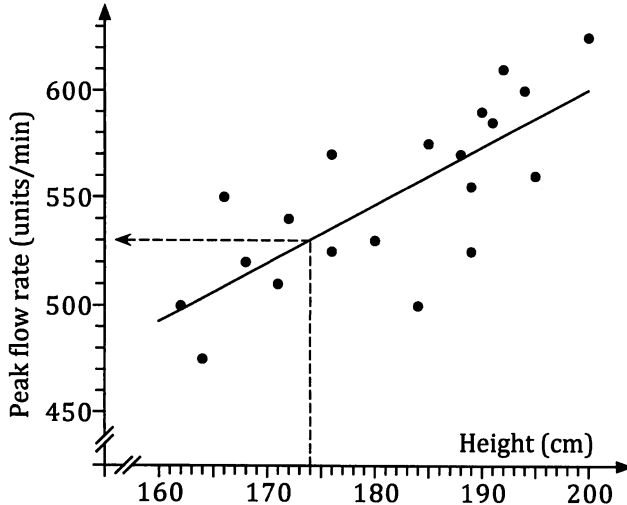


Estimated mark for absent student:
Approximately 17.5

Equation of line of best fit:
 $y = 0.39x + 7.4$

where x is the mark in the first test and y is the mark in the second test.

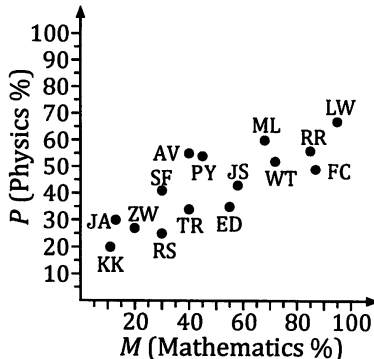
5.



Estimated peak flow for male aged between 30 and 50 and of height 174 cm is approximately 530 units/min.

Equation of line of best fit:
 $P = 2.66h + 67$

6.

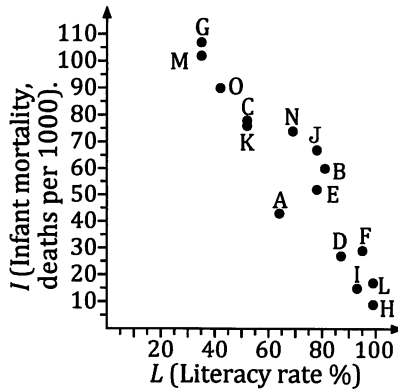


Graph indicates that in general those scoring high marks in one subject also scored high in the other, and similarly for those scoring low marks. Given the number of points involved and the reasonable linear correlation a linear model is suitable.

$P = 0.425M + 22$

Estimated physics mark: 49.

7. (a)

(b) $I = -1.32L + 149.4$

(c) (i) 86.

The graph indicates a good linear correlation so a linear model is suitable and the prediction involves *interpolation* so should be quite a reliable prediction.

(ii) 130.

This time *extrapolation* is involved with a literacy rate of 15% being well below the given literacy rates (for which the lowest is 35%). Hence prediction should not be relied upon.

Exercise 2B. Page 56.

1. No we cannot conclude that the reading of fashion magazines is causing the eating disorder. The fact that there is an observed association between the number of hours spent reading teen fashion magazines and the chance of having an eating disorder does not mean that one causes (or doesn't cause) the other.

What might be more likely to be happening here is that a person's level of concern about their own body image could be a confounding, or lurking, variable. If a person is overly concerned with their own appearance and body image this concern could drive them to look through teen fashion magazines to view pictures of other young people, and could also drive them into eating disorders as they seek their desired look.

2. No we cannot conclude that the duration of employment of the individual is causing the decline in company profits. The fact that there is an observed association between two variables does not mean that one necessarily causes the other. It is more likely in this situation that it is purely coincidental that the individual happened to begin employment at a time when company profits started their decline.

3. No we cannot conclude that it is the extra time at school that is causing the rise in the number of school leavers with attention deficit disorder. The fact that there is an observed association between two variables does not mean that one necessarily causes the other.

The confounding variable could well be time. With time the population will increase and so we might expect the number of cases of particular disorders to increase. As society evolves over time greater importance is attached to education, hence the increasing time at school and the greater number of years when learning order diagnoses could occur. Also, with time, greater awareness of learning disorders and better methods of diagnosing them develop, hence the rise in the number diagnosed with attention deficit disorder.

4. (a) $r = 0.74$

(b) $y = 0.346x + 5.48$

(c) 15.2 extrapolation

6. (a) $r = -0.84$

(b) $y = -0.385x + 18.87$

(c) (i) 15.0 interpolation

(ii) 3.5 extrapolation

8. (a) $r = 0.70$

(b) $y = 0.095x + 2.56$

(c) 3.6 extrapolation

5. (a) $r = -0.87$

(b) $y = -0.640x + 17.74$

(c) 13.9 interpolation

7. (a) $r = 0.88$

(b) $y = 0.365x + 3.83$

(c) (i) 18.4 extrapolation

(ii) 10.0 interpolation

9. (a) $r = -0.88$

(b) $y = -0.663x + 46.7$

(c) (i) 32.1 interpolation

(ii) 15.5 interpolation

10. (a) $r = 0.87$

(b) $y = 37.8x + 239.2$

(c) (i) 432 interpolation

(ii) 587 extrapolation

(iii) 727 extrapolation

11. (a) $r = 1$

(b) $y = 0.8x - 11$

(c) (i) 57 interpolation

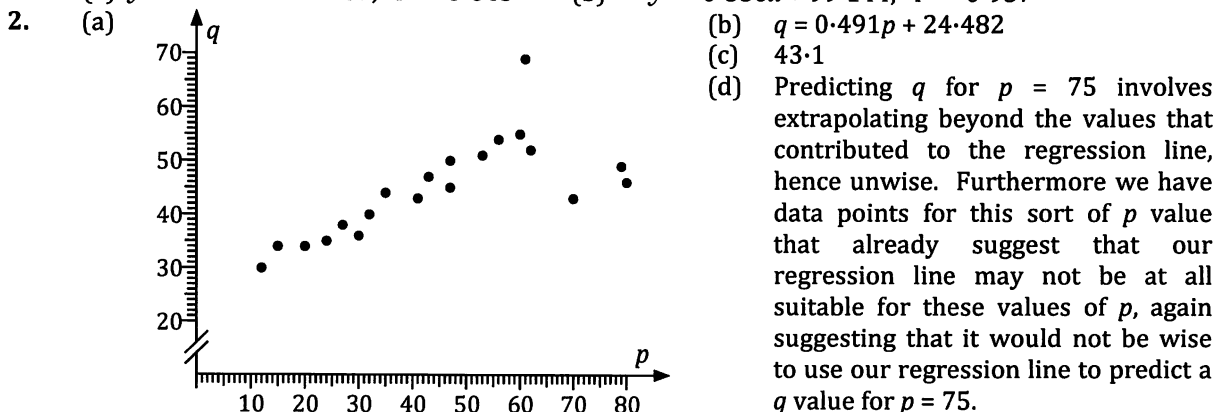
(ii) 21 interpolation

(iii) 149 extrapolation

12. (a) $q = -0.655p + 34.22$
 (b) $p = -1.199q + 46.24$
 (c) (i) 14.6
 (ii) 16.3
13. (a) $q = 0.2p + 12$
 (b) $p = 5q - 60$
 (c) (i) 21.6
 (ii) 100
14. (a) -0.73
 (b) $N = 399 - 8.38T$
 (c) (i) 181. Should be fairly reliable given r value and interpolation involved.
 (ii) 47. Extrapolation involved so validity is questionable.
15. (a) 0.83 (b) $N = 37.8 + 9.8T$
 (c) (i) 175. Should be fairly reliable given r value and interpolation involved.
 (ii) 8. Extrapolation involved so validity is questionable.
16. (a) 0.85 (b) End of year = $19.7 + 1.16 \times \text{mid year}$ (c) 93
17. (a) -0.97, $S = 35.3 - 0.38C$ (b) ~ 79%
18. (a) 0.86, $L = 52.1 + 0.20S$ (b) (i) 64 years (ii) 68 years
 (c) No we cannot conclude that a high life expectancy is due to having access to good sanitation. The data suggests there is a correlation between the two but we cannot conclude that one is due to the other. I.e. correlation does not allow us to assume cause and effect.

Exercise 2C. Page 65.

1. (a) $y = -0.366x + 87.467$, $r = -0.645$ (b) $y = -0.550x + 99.144$, $r = -0.957$



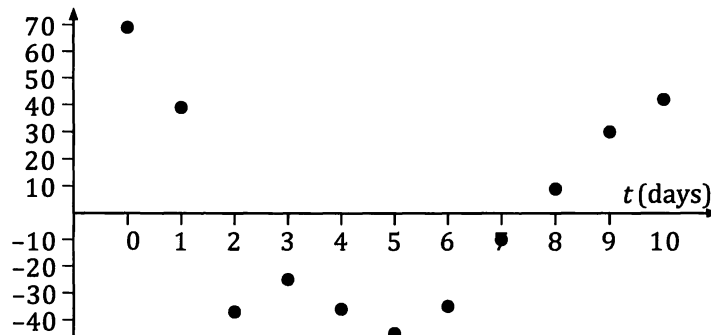
3. Table 1, Graph C. Table 2, Graph D. Table 3, Graph E.
 Table 4, Graph A. Table 5, Graph F. Table 6, Graph B.
4. According to the given information, approximately 64% ($0.802^2 \approx 0.64$) of the variation in the weights of men of the particular age can be explained by the variation in their heights.
5. Approximately 53% ($0.73^2 \approx 0.53$) of the variation in reading age can be explained by the variation in foot length.
 The given information does not mean that increasing the length of a person's foot causes their reading age to increase. Correlation shows association between variables but does not prove that the association is causal. Amongst primary school children we would expect reading age to generally increase with increasing foot size because the increased foot size will often mean an older child is being considered, and we would expect the older children to generally have higher reading ages. Hence the association is not a surprise but it does not prove cause and effect. In this case the "lurking variable", or "confounding variable", is the age of the primary school child.
6. (Graph not shown here.)
 $r = 0.51$ (correct to two decimal places),
 $r^2 = 0.26$ (correct to two decimal places).
 The value of r informs us that x and y have a moderate positive linear association.
 The value of r^2 informs us that 26% of the variation in the variables is explained by the linear association between them.

7. Viewing the scattergraph on a calculator or computer confirms that there seems to be an association between *units of fertiliser added* and *mean height of the seedlings after six weeks* that could well suit a linear model. (This is further confirmed by a correlation coefficient of 0.9736 indicating a very strong linear correlation.)

With the correlation coefficient, r , equal to 0.9736, $r^2 \approx 0.95$. Hence approximately 95% of the variation in seedling height can be accounted for by the variation in the number of units of fertiliser added.

8. (a)

t days	0	1	2	3	4	5	6	7	8	9	10
Residual	69	39	-37	-25	-36	-45	-35	-10	9	30	42



- (b) Residuals are not random. Each end of the table is giving positive residuals, negative residuals occurring in the middle of the table. This suggests that using this linear model to predict P for higher t values will increasingly underestimate the P value.

9. (a)

h	1	2	3	4	5	6	7	8	9	10
Residual	245	82	-41	-124	-162	-165	-123	-41	81	243

- (b) Residuals are not random. Suggests linear model may not be the most appropriate.

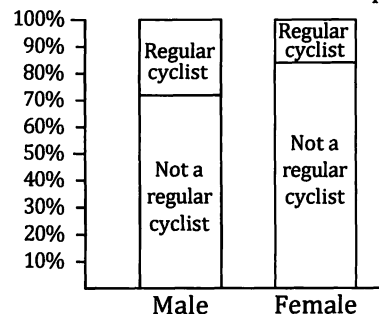
(c) $a = 20.4$, $b = 8.5$, $c = -13.9$.

- (d) $r \approx 4700$. Viewing the best fit quadratic on a calculator shows an extremely good fit to the given data points. However the prediction involves extrapolation so, without further knowledge, the predicted value should not be regarded as being particularly reliable, especially as we are extrapolating quite a long way from measured h values.

Miscellaneous Exercise Two. Page 70.

- No we would not be correct to conclude that the firemen were causing the damage. Correlation does not prove causation. The fact that A and B are correlated does not of itself prove any causal relationship between them. In the given situation a more sensible explanation is that the correlation is due to some other factor, such as the size of the fire. The larger the fire the more damage we would expect to be done and the more firemen likely to be in attendance. Thus size of fire is likely to be the lurking, or confounding, variable.
- We wish to see if the proportion who are regular cyclists changes as we move across the gender boundary. Are the proportions in any way explained by gender? Thus we need column percentages. Or, using the rule, the explanatory variable is gender and this uses columns to display its categories so we want column percentages.

	Male	Female
Regular cyclist	28%	16%
Not a regular cyclist	72%	84%
Totals	100%	100%

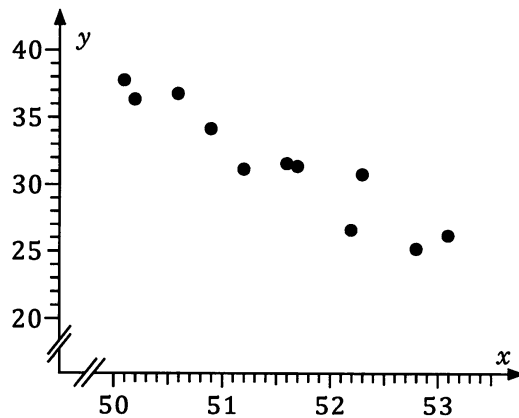


For the group interviewed the majority of both the males and the females were not regular cyclists (72% of the males and 84% of the females). However this does mean that of the males interviewed 28% were regular cyclists compared to 16% of the females.

Thus as we move across the gender boundary the proportion of regular cyclists does show quite a change.

The research data collected does suggest that there is an association between the likelihood of someone being a regular cyclist and gender. If there were no association we would expect little change in the percentage of regular cyclists as we move across the gender boundary. This is not the case with the given data.

3. (The predicted values given here are the values given by the least squares regression line. Your "eyeballed" values should be reasonably close to these values.)



(a), (b) and (c):

The scattergraph indicates that a linear model would be an appropriate choice and should give reliable predictions when interpolation is involved.

- (a) When $x = 50.8$ the linear model suggests $y \approx 34.5$. The suitability of the model and the fact that interpolation is involved means that this predicted y value should be very reliable.
 (b) When $x = 53.2$ the linear model suggests $y \approx 24.7$. This prediction involves extrapolation but 53.2 is not far beyond the given x values so the prediction should still be viewed as quite reliable.
 (c) When $x = 58$ the linear model suggests $y \approx 5$ but $x = 58$ involves extrapolation far beyond the given x values. Hence the predicted y value should not be considered at all reliable.

4. (a) $r_{AC} = 0.82$, $r_{BC} = 0.50$. Test A is better indicator. (b) 70 (c) 1, 3, 8, 9, 11, 15.

5. (a) The test B marks were generally higher than the test A marks, with B having a mean score of 37.77 and a median of 39, and A a mean score of 32.43 and a median of 31. Twenty one of the thirty students achieved a higher mark in test B than they did in test A, two students scored the same and seven scored more in A than in B.

Both sets of scores had a low score of 17 and a high score of 49 and hence the sets of scores had the same range (32). However in set B, two scores lie noticeably further away from the rest so whilst the range of the two sets is the same, other measures of spread indicate that the set B scores are less spread out than the set A scores. The interquartile range is 15 for set A and 9 for set B and standard deviation is 9.29 for set A and 7.20 for set B (9.45 and 7.32 if using σ_{n-1}).

If the two outliers are removed from the test B scores this difference in spread is even more noticeable.

- (b) The shape of the scattergraph suggests that linear regression is appropriate and the score A mark of 27 achieved by the 31st student means interpolation will be involved, further justifying the validity of regression analysis. Cropping the data so that the two outlying points are excluded, and hence do not unduly influence the regression line, gives a regression line:

$$\text{Score B} = 0.497 \times \text{Score A} + 22.70$$

and a predicted mark of 36.1 in B.

With 28 points involved, good linear correlation ($r = 0.883$) and interpolation used, the estimate is likely to be reasonably valid. However it is only an estimate and, as two students did show, it is quite possible to score a lower mark in test B than the test A mark would generally suggest.

(With the two outlying scores included the regression line is

$$\text{Score B} = 0.611 \times \text{Score A} + 17.94 \quad \text{to give a predicted mark of } 34.4.)$$

Exercise 3A. Page 73.

1. 13, 15.
2. 320, 640.
3. 14, 16.
4. G, H.
5. S, V.
6. VII, VIII.
7. 22, 25.
8. J, A.
9. 13, 21.
10. 49, 64.
11. 17, 19.
12. 28, 36.
13. 0.00001, 0.000001.
14. 64, 128.
15. 729, 2187.
16. 79, 72.
17. 3412, 4123.
18. 73, 84.
19. 0.015625, 0.0078125.
20. $0.\overline{142857}$, 0.125.
21. 127, 255.
22. -1, -3.
23. 364.5, 546.75.
24. $823543 (= 7^7)$, $16777216 (= 8^8)$.

Exercise 3B. Page 75.

1. 14
2. 17
3. 31
4. 42
5. 44
6. 28
7. 39
8. 23
9. 32
10. 3
11. 3
12. 3
13. 7
14. 255
15. 262
16. 21
17. 30
18. 157
19. 23
20. 16
21. 511
22. 4
23. 8
24. 16
25. 26
26. 15
27. 17
28. 32
29. 3
30. 5
31. 7
32. 65
33. 1
34. 3
35. 2
36. 3
37. 5
38. 42
39. 0
40. 55
41. 1
42. 7
43. 4
44. 7
45. 11
46. 94
47. 0
48. 123

Exercise 3C. Page 81.

1. a I, b A, c B, d D, e G, f K, g H, h D, i F.
2. 11, 16, 21, 26, 31, 36.
3. 4, 3, 1, -3, -11, -27.
4. 3, 7, 11, 15, 19.
5. 1, 4, 7, 10, 13.
6. 2, 5, 11, 23, 47.
7. 2, 5, 14, 41, 122.
8. 1, 3, 13, 63, 313.
9. $T_1 = 128, T_2 = 160, T_3 = 200, T_4 = 250, T_{n+1} = 1.25 \times T_n$.
10. $T_1 = 1000, T_2 = 1200, T_3 = 1440, T_4 = 1728, T_{n+1} = 1.2 \times T_n$.
11. $T_1 = 2000, T_2 = 1700, T_3 = 1445, T_4 = 1228.25, T_{n+1} = 0.85 \times T_n$.
12. $T_1 = 5, T_{n+1} = T_n + 4$.
13. $T_1 = 26, T_{n+1} = T_n - 3$.
14. $T_1 = 6, T_{n+1} = 2T_n + 1$.
15. $a = 4, T_3 = 24$.
16. 15, 23, 31, 39, 47, 55.
17. 10, 26, 74, 218, 650, 1946.
18. 12, 23, 45, 89, 177, 353.
19. 1, 4, 5, 9, 14.
20. 7, 7, 14, 21, 35.
21. 3, 4, 2, 6, -2.
22. 1, 2, 3, 6, 11, 20.
23. $a = 5, b = 3, T_6 = 5468$.
24. $a = 2, b = -3, T_6 = 67$.
25. 69, 56, 47, 41, 37.25.
26. 1, -2, -4, 8, 16.
27. 3, 6, -12, -24, 48.
28. 3, 9, 144, 18000, 23328000.

29. (a)

Stacking Cans					
T_1	T_2	T_3	T_4	T_5	T_6
1	3	6	10	15	21

(b) IV

Spreadsheets. Page 84.

①.

	A	B	C	D
1	1	6	5	1
2	2	7	10	4
3	3	8	15	9
4	4	9	20	16
5	5	10	25	25
6	6	11	30	36
7	7	12	35	49
8	8	13	40	64

②. Square

③. Yes:

A	B	C	D
n	n+5	5n	?

$$\begin{aligned}
 ? &= n(n+5) - 5n \\
 &= n^2 + 5n - 5n \\
 &= n^2, \text{ as required.}
 \end{aligned}$$

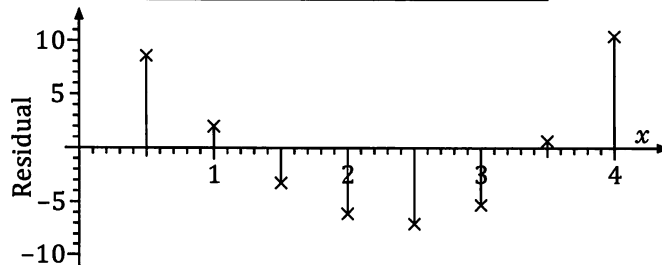
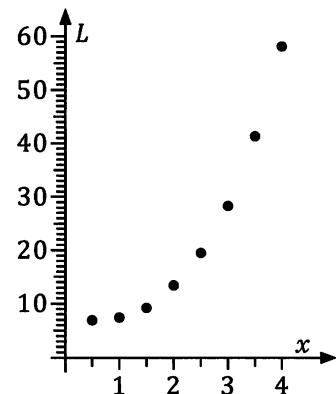
Miscellaneous Exercise Three. Page 84.

- (a) 1093, 3280. (b) 343, 512. (c) 262·144, 209·7152. (d) -200, -400. (e) $\frac{7}{8}$, 1.
- 1, 2, 9, 58.
- 120, 58, 27, 11·5.
- The 3·1 suggests that in general, if the consumption of the foodstuff in a country goes up by 1 kilogram per person per year we would expect to see the death rate due to heart disease for that country increase by approximately 3·1 deaths per year per 100 000 of population.
The 90·9 suggests that if a country has a zero consumption of the foodstuff the death rate due to heart disease would be approximately 90·9 deaths per year per 100 000 of population. (However, zero consumption of the foodstuff is outside the range of consumption figures used to form the line of best fit so this suggested death rate due to heart disease figure should not be relied upon.)
- $a = 11$, $T_3 = 278$.
- 20, 42, 40, 14, -36.

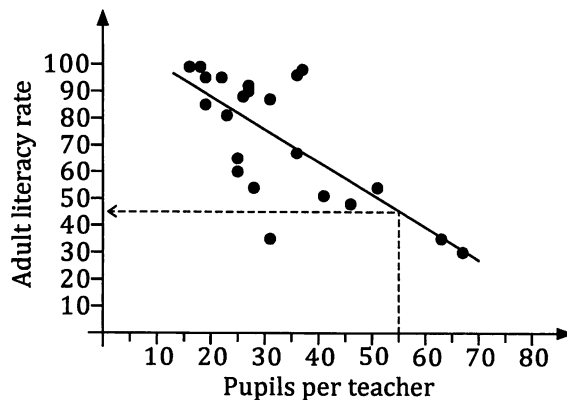
- The student's conclusion is incorrect. The data is not best suited to linear modelling. Had the student viewed a scattergraph of the data, see right, it should have been apparent that a linear model was probably not the most appropriate but that an exponential or perhaps a polynomial function would have been better.

A look at the residuals or a residual plot, see below (residuals calculated using equation given in question), should similarly have alerted the student to this fact as both show there is a pattern to the residuals and for linear modelling to be appropriate this should not be the case.

x	0·5	1·0	1·5	2·0	2·5	3·0	3·5	4·0
Residual	8·55	2	-3·25	-6·1	-7·05	-5·3	0·65	10·4



- (a)



- No. Data suggests there is a correlation between literacy rate and pupils per teacher but cannot conclude one is due to the other.

- Approx 45%

9. (a) For the line $y - \bar{y} = \frac{s_{xy}}{s_x^2}(x - \bar{x})$ to pass through $(\bar{x}, \bar{y})$ the values $x = \bar{x}$ and $y = \bar{y}$ must "fit"

the equation. Substituting $x = \bar{x}$ gives $y - \bar{y} = \frac{s_{xy}}{s_x^2}(\bar{x} - \bar{x})$

$$\text{Hence} \quad y - \bar{y} = 0$$

$$\text{i.e.} \quad y = \bar{y}$$

Thus the least squares regression line passes through $(\bar{x}, \bar{y})$.

- (b) Because it is the line that *minimises* the sum of the *squares* of the residuals.

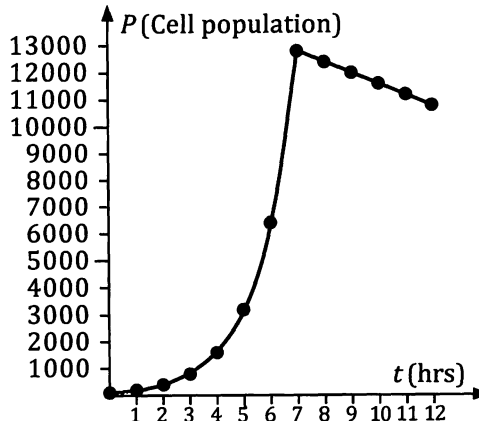
10. Allow others to read and comment upon your report and similarly you read and comment upon the reports of others.

Exercise 4A. Page 93.

1. $T_1 = 6, T_{n+1} = T_n + 4.$
 2. $T_1 = 28, T_{n+1} = T_n - 2.$
 3. $T_1 = 5, T_{n+1} = T_n + 10.$
 4. $T_1 = 7.5, T_{n+1} = T_n + 2.5.$
 5. $T_1 = 100, T_{n+1} = T_n - 11.$
 6. $T_1 = 6, T_n = 2T_{n-1}.$
 7. $T_1 = 0.375, T_n = 4T_{n-1}.$
 8. $T_1 = 384, T_n = 0.25T_{n-1}.$
 9. $T_1 = 50, T_n = 3T_{n-1}.$
 10. $T_1 = 1000, T_n = 1.1T_{n-1}.$
 11. Neither arithmetic nor geometric.
 12. Geometric.
 13. Arithmetic.
 14. Arithmetic.
 15. Neither arithmetic nor geometric.
 16. Geometric.
 17. Geometric.
 18. Arithmetic.
 19. Neither arithmetic nor geometric.
 20. Neither arithmetic nor geometric.
 21. Arithmetic.
 22. Geometric.
 23. $T_1 = 8, T_2 = 11, T_3 = 14, T_4 = 17, T_{n+1} = T_n + 3.$
 24. $T_1 = 100, T_2 = 97, T_3 = 94, T_4 = 91, T_{n+1} = T_n - 3.$
 25. $T_1 = 11, T_2 = 22, T_3 = 44, T_4 = 88, T_{n+1} = 2T_n.$
 26. $T_1 = 2048, T_2 = 1024, T_3 = 512, T_4 = 256, T_{n+1} = 0.5T_n.$
 27. (b) $N_{n+1} = N_n + 800$
 28. (a) The sequence is a geometric progression.
(b) The next three terms after the first are 550, 605 and 665.5.
 29. (a) The sequence is a geometric progression.
(b) The next three terms after the first are 1250, 1562.5 and 1953.125.
 30. (a) The sequence is a geometric progression.
(b) The next three terms after the first are 21600, 19440 and 17496.
 31. (a) $T_1 = 3, T_{n+1} = T_n + 5.$ (b) The sequence is arithmetic.
 32. (a) $T_1 = 1.5, T_{n+1} = T_n \times 2.$ (b) The sequence is geometric.
 33. (a) $T_1 = 4, T_2 = 9, T_3 = 16, T_4 = 25, T_5 = 36.$ (b) Neither arithmetic nor geometric.
 34. (a) After 1 year, 2 years, 3 years and 4 years the account is worth \$1 296, \$1 392, \$1 488 and \$1 584 respectively.
(b) The amounts are in arithmetic progression. (c) $T_1 = \$1200, T_{n+1} = T_n + \$96.$
 35. $T_1 = \$45000, T_{n+1} = T_n + \$1500.$ The terms of the sequence progress arithmetically.
 36. \$68 000 in 2014, \$71 400 in 2015, \$74 970 in 2016, \$78 718.50 in 2017.
 $T_1 = \$68\,000, T_{n+1} = 1.05T_n.$
 37. $T_1 = \$1\,500, T_{n+1} = 1.08T_n.$
 38. (a) \$21.00 (b) \$37.90 (c) \$54.80
 39. \$124 000, \$110 000, \$96 000.
 40. \$22 750, \$18 500, \$14 250.
- After 70000 copies in its first year the value of the copier will be \$26405.
41. $T_1 = \$36\,000, T_{n+1} = 0.85T_n.$

Exercise 4B. Page 103.

1. $T_{100} = 506$
2. $T_{100} = 289$
3. $T_{100} = 815$
4. $T_{100} = -120$
5. $T_{25} = 5 \times 2^{24}$
6. $T_{25} = 1.5 \times 4^{24}$
7. $T_{25} = 8 \times 3^{24}$
8. $T_{25} = 11 \times 2^{24}$
9. $T_{28} = 223$
10. $T_{20} = 3\,495\,265$
11. $T_{19} = 774\,840\,977$
12. $T_{45} = 6$
13. $T_1 = 48$, $T_{n+1} = T_n + 3$. Julie successfully completes 90 items on the 15th day.
14. Substituting y for T_n and x for n the rule $T_n = a + (n - 1)d$ becomes $y = dx + (a - d)$. This is the equation of a straight line with gradient d , cutting the y -axis at $(0, a - d)$.
15. Substituting y for T_n and x for n the rule $T_n = ar^{n-1}$ becomes $y = ar^{x-1}$, i.e. $y = \left(\frac{a}{r}\right)r^x$. An exponential function cutting the y -axis at $\left(0, \frac{a}{r}\right)$.
16. As $n \rightarrow \infty$ the "nd" term in the expression $a + (n - 1)d$ will dominate. Thus as $n \rightarrow \infty$, T_n will be increasingly large and positive if $d > 0$ and increasingly large and negative if $d < 0$.
17. As $n \rightarrow \infty$ the n in the expression ar^{n-1} will dominate. Thus as $n \rightarrow \infty$, if $r > 1$, T_n will become increasingly large, either positively or negatively dependent on the sign of the constant a .
if $r < -1$, T_n will become increasingly large, alternating between large negative and large positive.
if $-1 < r < 1$, T_n will become smaller and smaller, maintaining the sign of the constant a if r is positive and alternating between small positive and small negative if r is negative. I.e., if $-1 \leq r \leq 1$, as $n \rightarrow \infty$, $T_n \rightarrow 0$.
18. The first four terms are 8, 11, 14, 17. The 50th term is 155. The 100th term is 305.
19. The first four terms are 100, 97, 94, 91. The 50th term is -47. The 100th term is -197.
20. 11, 22, 44, 88, 180 224, 184 549 376.
21. 2048, 1024, 512, 256, 0.0625.
22. (a) $T_n = 9 + (n - 1) \times 6$, i.e. $T_n = 6n + 3$. (b) $T_n = 7 + (n - 1) \times 1.5$, i.e. $T_n = 1.5n + 5.5$.
23. (a) $T_n = 3 \times 2^{n-1}$ (b) $T_n = 100 \times 1.1^{n-1}$
24. (a) 856 (b) 3495 (c) The 142 858th term.
25. (a) 126 953.125 (b) The 14th term.
26. (a) 844 700 (b) The 60th term.
27. 1, 8, 27, 64, neither.
28. (a) 64 (b) 7
29. (a) 1850 (b) 2 000
30. (a) 7 971 615 (b) 5
31. (a) -1 835 008 (b) 7
32. The amount in the account at the end of ten years is \$8635.70.
33. (a) 12800 (b) 10800 (c)
34. (a) \$48 000 (b) \$59 000
35. Approximately 61 million, assuming annual growth rate for the given years continues.
36. Approximately 6000.
37. (a) 1.2 m (b) Approximately 9 cm.
38. (a) \$3 130 000 (b) \$3 920 000 (c) Approximately fifteen and a half years.
39. Just after the end of the 22nd year, i.e. early in the 23rd year.

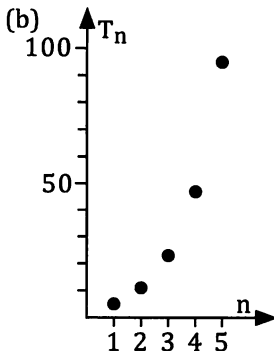


Exercise 4C. Page 108.

1. $T_1 = 10$, $T_2 = 23$, $T_3 = 49$, $T_4 = 101$.
2. $T_1 = 200$, $T_2 = 450$, $T_3 = 1200$, $T_4 = 3450$.
3. $T_1 = 200$, $T_2 = 150$, $T_3 = 50$, $T_4 = -150$.

4. Sequence A

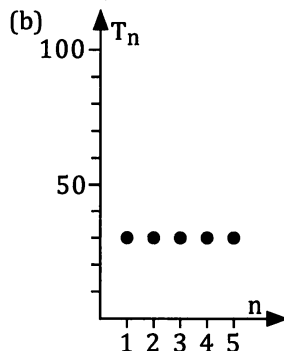
- (a) $T_1 = 5$,
 $T_2 = 11$,
 $T_3 = 23$,
 $T_4 = 47$,
 $T_5 = 95$.



(c) Increasing sequence.

Sequence B

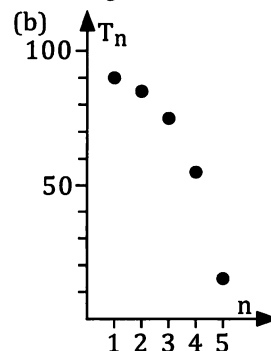
- (a) $T_1 = 30$,
 $T_2 = 30$,
 $T_3 = 30$,
 $T_4 = 30$,
 $T_5 = 30$.



(c) Steady state sequence.

Sequence C

- (a) $T_1 = 90$,
 $T_2 = 85$,
 $T_3 = 75$,
 $T_4 = 55$,
 $T_5 = 15$.



(c) Decreasing sequence.

5. The terms increase towards a value of 50000, getting closer and closer to 50000 as n becomes increasingly large.
 Multiplication by 0.96 produces a 4% decrease but we also add 2000. With 2000 being greater than 4% of 35000 the terms will increase in size but eventually the 4% decrease will almost balance the 2000 increase. The long term steady state referred to in the question will result as the decrease, 4% of T_n , approaches the increase, 2000, i.e. as T_n approaches 50000.
6. In 1 year, and just after the catch, the fish population would be approximately 1940.
 In 2 years, and just after the catch, the fish population would be approximately 1873.
 In 5 years, and just after the catch, the fish population would be approximately 1619.
 In 10 years, and just after the catch, the fish population would be approximately 947.
7. $T_n = 1.012 \times T_{n-1} + 120\,000$, $T_0 = 17\,600\,000$.
 $T_5 \approx 19\,300\,000$, i.e. approximately 19.3 million.
 $T_{10} \approx 21\,100\,000$, i.e. approximately 21.1 million.

Miscellaneous Exercise Four. Page 109.

1. (a) 243, 729.
 (b) O, U (vowels) or M, Q (alphabetical, jumping three letters each time).
 (c) S, S. (d) 0.5, 0.25.
2. (a) 7, 17, 37, 77, 157, 317. (b) 6, 11, 21, 41, 81, 161.
 (c) 256, 128, 64, 32, 16, 8. (d) 5, 12, 19, 26, 33, 40.
3. (a) 43 (b) 22 (c) 6 (d) 8
 (e) $a = 4$, $b = -1$, $T_9 = 43\,691$, $T_{10} = 174\,763$.
4. 157 5. 0.1875
6. The 0.9 suggests that in general, if we take two people from this group, for each 1 cm difference in height we would expect to find the taller person weighing approx 0.9 kg more than the shorter.
7. $r_{xy} = 0.86$, $y = 29.5 + 0.532x$ (a) 46.5 (b) 61.4.

A scattergraph of the data, and the value of r for the ten data points, both indicate that it is valid to use linear regression to predict. However, we must view the part (b) prediction with great caution as extrapolation is involved with $x = 60$ quite a way beyond the x values ($22 \leq x \leq 49$) in the given data.

8. (a) From the first term value of 5000 subsequent terms decrease, getting closer and closer to 4000 as n gets bigger and bigger. (Because as T_n gets closer to 4000 the 20% reduction is almost exactly balancing the 800 that is added.)
 (b) From the first term value of 5000 subsequent terms increase, getting closer and closer to 8000 as n gets bigger and bigger. (Because as T_n gets closer to 8000 the 20% reduction is almost exactly balancing the 1600 that is added.)
 (c) From the first term value of 5000 subsequent terms become bigger and bigger with $T_n \rightarrow \infty$ as n becomes increasingly large. (Because from the start the increase of 20% exceeds the 800 we are taking away so subsequent terms will increase "without bound".)
9. $T_{n+1} = 1.08 \times T_n - \200 , $T_{10} = \$5738.39$.
10. (Graph not shown here.) Applying a linear model suggests that a country with 285 motor vehicles per 1000 people would have approximately 385 (nearest 5) televisions per 1000 people. With eighteen points involved, the distribution of those points indicating a linear model to be appropriate, $r = 0.936$ and interpolation being involved we can have reasonable confidence in our predicted value.
 Approximately 88% ($r^2 = 0.876$) of the variation in numbers of televisions can be explained by the variation in the number of motor vehicles.

11.

	Never	Once per week	Twice per week	Three times per week	Four times per week	Five times per week
Male	24%	10%	12%	18%	20%	16%
Female	24%	8%	14%	15%	21%	18%

As we move across the gender categories the proportions in each frequency of engagement in sport or exercise change very little. For both males and females the once a week attendance produced the lowest percentages of 10% and 8% respectively. For both gender categories 24% of those surveyed said that in an average week they never engage in sport or exercise.

The figures suggest that there is no association between gender and the frequency with which an adult engages in sport or exercise.

Exercise 5A. Page 113.

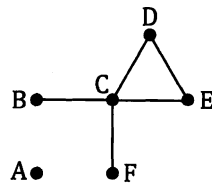
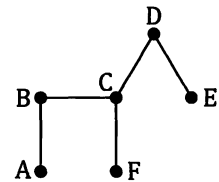
- (a) From Kellerberrin to Merredin is 58 km.
 (b) From Northam to Coolgardie is 461 km.
 (c) The entire journey would take 7 hours and 15 minutes.
- ABDF 237 km, ABCDF 264 km, ACDF 213 km, ACEF 225 km, AEF 215 km.
- The cheapest the system can be done for is \$3990.
- Vertices 5, Edges 5, Faces 2. 5. Vertices 5, Edges 8, Faces 5.
- Vertices 8, Edges 8, Faces 2. 7. Vertices 2, Edges 1, Faces 1.
- Vertices 5, Edges 6, Faces 3. 9. Vertices 5, Edges 8, Faces 5.
- Vertices 5, Edges 8, Faces 5. 11. Vertices 7, Edges 10, Faces 5.
- Vertices 1, Edges 2, Faces 3. 13. Vertices 5, Edges 7, Faces 4.
- Vertices 7, Edges 8, Faces 3. 15. Vertices 7, Edges 9, Faces 4.
- Vertices 5, Edges 9, Faces 6. 17. Vertices 7, Edges 12, Faces 7.
- Vertices 6, Edges 9, Faces 5. 19. Vertices 6, Edges 9, Faces 5.
- Vertices 10, Edges 13, Faces 5. 21. Vertices 12, Edges 16, Faces 6.

Exercise 5B. Page 128.

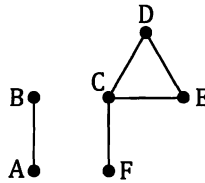
- (a) Connected (b) Connected (c) Disconnected (d) Disconnected (e) Connected
- (c) and (e) are simple graphs.
 (a) is not a simple graph because it contains multiple edges.
 (b) is not a simple graph because it contains a loop.
 (d) is not a simple graph because it is directed.
 (f) is not a simple graph because it is weighted.
- (a) Length 5 (b) Length 3 (c) Length 6

4. (a) Digraph. A might like B but B does not necessarily like A. Hence the need to show direction.
 (b) Digraph. Just because A is aware of the existence of B it does not necessarily follow that B is aware of the existence of A. For example suppose B is someone famous. A may well be aware of the existence of B without B being aware of the existence of A. Hence the need for direction.
 (c) Undirected. No need to show direction because if A is accepted as a Facebook friend of B then B is automatically a Facebook friend of A.
 (d) Undirected. No need to show direction. If A is related to B then B must be related to A.
 (e) Undirected. No need to show direction. If A has been to the movies with B then B must have been to the movies with A.
 (f) Undirected. No need to show direction. If A lives within 5 km of B then B must live within 5 km of A.
 (g) Digraph. Just because A has the phone number of B it does not necessarily follow that B must have the phone number of A. Hence the need to show direction.

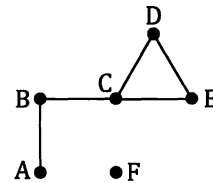
5. (a) If the edge CE is removed the graph remains connected, see diagram on the right. Hence CE is not a bridge.
 (b) In the connected network given in the question, edges AB, BC and CF are bridges because, in each case, removing the edge leaves the network unconnected, as shown below.



Edge AB removed.
Graph unconnected.

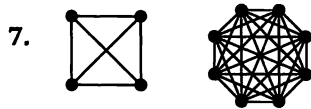


Edge BC removed.
Graph unconnected.

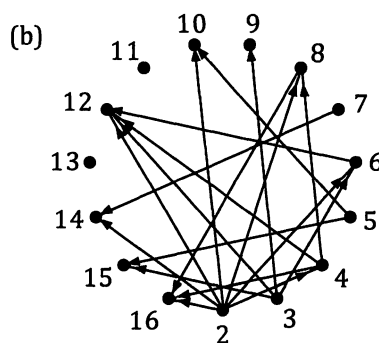
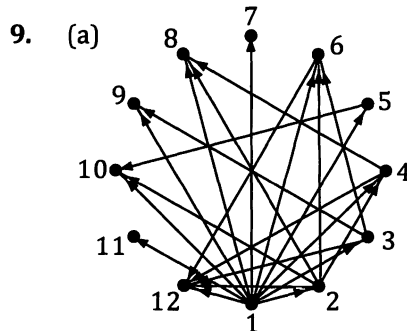


Edge CF removed
Graph unconnected.

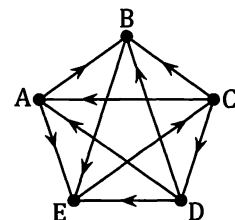
6. Only (c) and (d) are complete graphs.



8. (a) Of the given sequences of vertices the walks are: ABCDE, BCD, ABDEF, ABDEFBC, BDEFB.
 (b) ABCDE: A path. BCD: A path. ABDEF: A path.
 ABDEFBC: A trail. BDEFB: A cycle

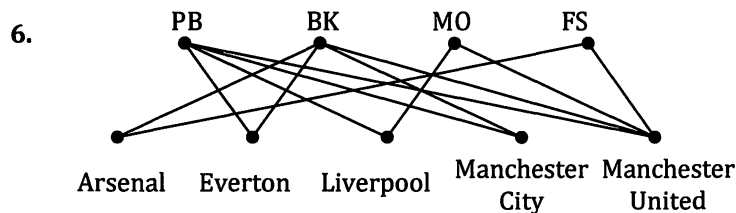


10.

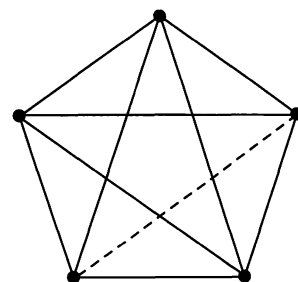


Exercise 5C. Page 134.

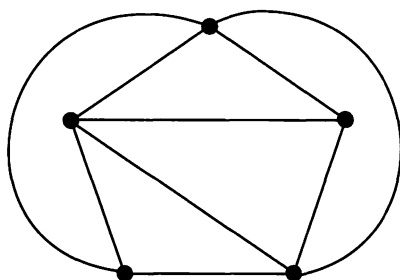
1. The network has 6 edges.
2. The network has 4 vertices.
3. The network has 9 faces.
4. (a) A cube has 8 vertices, 6 faces, 12 edges. Euler's rule is obeyed.
 (b) A regular tetrahedron has 4 vertices, 4 faces, 6 edges. Euler's rule is obeyed.
 (c) A regular octahedron has 6 vertices, 8 faces, 12 edges. Euler's rule is obeyed.
5. Only graph (c) is a subgraph of the given graph.
 Graph (a) is not because it contains the edge FE which is not in the original.
 Graph (b) is not because it contains the edge CD which is not in the original.



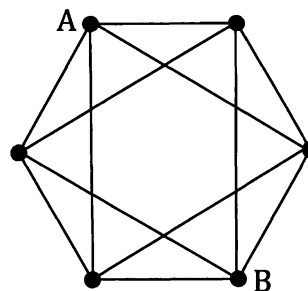
7. (a) If the broken line were to be added to the graph, as shown on the right, the graph would be complete because every vertex would be joined to every other vertex by an edge. However without this line there are two vertices that are not joined by an edge and hence the graph as presented in the question is not complete.



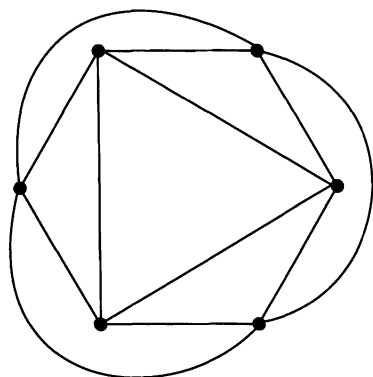
(b)



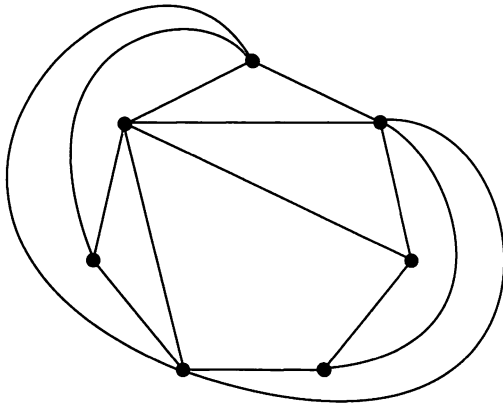
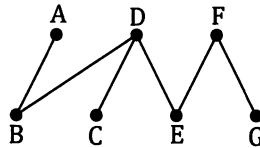
8. (a) The graph is redrawn on the right with two vertices labelled A and B. These two vertices are not joined by a single edge. Hence the graph is not a complete graph. (The vertices labelled A and B are not the only pair of vertices that could be chosen to make this point.)



(b)



9.

10. $\{A, D, F\}$ and $\{B, C, E, G\}$ 

11. No the given graph is not a bipartite graph.

Suppose we were to split the vertices into two groups. Vertex A would have to be in one of the groups and so, with A having an edge to vertex B and an edge to vertex C, B and C must be in the other group if A is not to be joined to a vertex in its own group. However B and C would then be in the same group as each other and are joined by an edge. Hence not every edge would join a vertex from one group to a vertex of the other. Hence the graph is not bipartite.

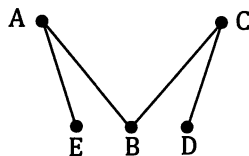
12. Yes the given graph is a bipartite graph.

Divide the vertices into two groups, $\{A, C\}$ and $\{B, D\}$. The given graph involve four edges, AB, BC, CD and DA. Each one of these joins a vertex from one of these two groups to a vertex from the other. Hence the graph is bipartite.

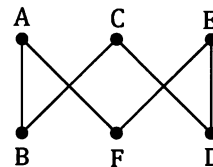
13. (a) The graph is not bipartite.

To be bipartite we must be able to split the vertices into two groups with every pair of adjacent vertices having one vertex from each group. Suppose that whatever group we put vertex A in we call group 1 and the other group we call group 2. An edge joins A to B and an edge joins A to E. Hence for the graph to be bipartite vertices B and E must be in group 2. But then C, being joined to B, and D, being joined to E, must each be in group 1. But then C and D are joined but in the same group. Hence the graph cannot be bipartite.

(b) The graph is bipartite.



(c) The graph is bipartite.



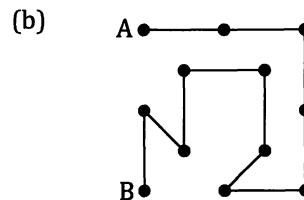
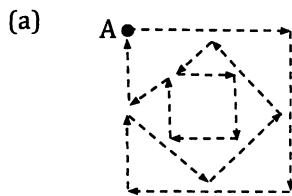
14. (a) The 3 graphs are complete graphs because each is a simple graph and in each one every pair of vertices are adjacent vertices. I.e. there is a path from each vertex in the graph to each of the other vertices in the graph that passes along just one edge.

(b) No it cannot be done for the other two graphs.

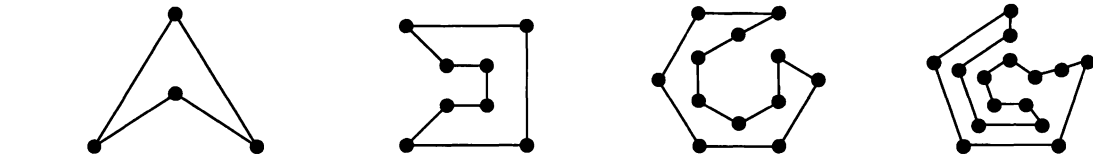
15. Compare your answer with that of others in your class and read about this classic network problem on the internet by searching for *The Water, Gas and Electricity Problem* or *The Three Utilities Problem*.

Exercise 5D. Page 141.

1. A degree 3. B degree 4. C degree 1. D degree 3. E degree 3. F degree 4.
 G degree 1. H degree 3. I degree 4. J degree 3. K degree 2. L degree 5.
 M degree 4. N degree 2.
2. A in-degree 2, out-degree 2. B in-degree 2, out-degree 1.
 C in-degree 1, out-degree 2. D in-degree 1, out-degree 2.
 E in-degree 2, out-degree 2. F in-degree 3, out-degree 2.
 G in-degree 2, out-degree 2. H in-degree 1, out-degree 1.
 I in-degree 1, out-degree 2. J in-degree 1, out-degree 1.
 K in-degree 2, out-degree 1.
3. (a) Semi-Eulerian. (b) Neither Eulerian nor semi-Eulerian.
 (c) Eulerian. (d) Semi-Eulerian.
 (e) Semi-Eulerian. (f) Eulerian.
4. (a) Many possible correct answers, one of which is ABCAFEDCEA.
 (b) Many possible correct answers, one of which is ABCDEFA.
5. (a) The graph is neither Eulerian nor semi-Eulerian. (Four odd vertices.)
 (b) The graph is semi-Hamiltonian.
6. (The answers shown below are not the only possible answers)



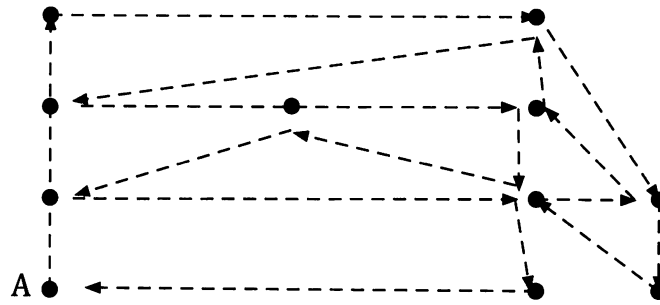
7. (The answers shown below are not the only possible answers)



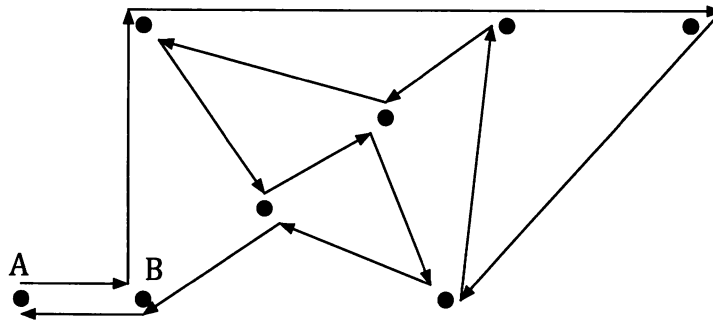
- 8.
- Eulerian Graph Semi Eulerian Graph Hamiltonian Graph Semi Hamiltonian Graph

- 9.
- Eulerian Graph Semi Eulerian Graph Hamiltonian Graph Semi Hamiltonian Graph

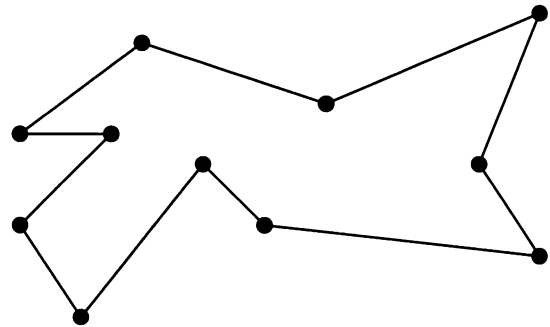
10. One possible route is:



11. The graph is not an Eulerian graph so we cannot find an Eulerian circuit starting at A and finishing at A. However there are semi-Eulerian trails starting at A and finishing at B, where B is as shown in the diagram below. The route should follow one of these trails and then the road between B and A can be travelled for a second time to return to A.



12. One of many possible circuits is shown on the right.



Exercise 5E. Page 146.

1.
$$\begin{array}{c} A \\ B \\ C \\ D \end{array} \begin{bmatrix} A & B & C & D \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

2.
$$\begin{array}{c} A \\ B \\ C \\ D \end{array} \begin{bmatrix} A & B & C & D \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 2 & 0 & 1 & 0 \end{bmatrix}$$

3.
$$\begin{array}{c} A \\ B \\ C \\ D \\ E \end{array} \begin{bmatrix} A & B & C & D & E \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

4.
$$\begin{array}{c} A \\ B \\ C \\ D \end{array} \begin{bmatrix} A & B & C & D \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

5.
$$\begin{array}{c} A \\ B \\ C \\ D \end{array} \begin{bmatrix} A & B & C & D \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{bmatrix}$$

6.
$$\begin{array}{c} A \\ B \\ C \end{array} \begin{bmatrix} A & B & C \\ 0 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$7. \quad K_3 \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad K_4 \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$8. \quad \begin{matrix} A \\ B \\ C \\ D \\ E \\ F \\ G \\ H \end{matrix} \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

9. (a) 3. They are ADAC, ACBC, ACAC.
 (b) 3. They are ADAB, ABCB, ACAB.
 (c) Zero.
 (d) 2. They are BCBC, BCAC.

$$M = \begin{matrix} & A & B & C & D \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$M^3 = \begin{matrix} & A & B & C & D \\ \begin{matrix} A \\ B \\ C \\ D \end{matrix} & \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 2 & 1 & 0 \\ 2 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$10. \quad (a) \quad X = \begin{matrix} & A & B & C & D & E \\ \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$(b) \quad X + X^2 = \begin{matrix} & A & B & C & D & E \\ \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix} & \begin{bmatrix} 2 & 2 & 2 & 1 & 1 \\ 2 & 2 & 2 & 1 & 1 \\ 2 & 2 & 4 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$

Matrix X shows the single flight connections between two of the cities.

Matrix X^2 shows the connections that can be done in two flights.

Thus if there are no zeros in matrix $X + X^2$ then every one of the five cities can be connected to each of the others by taking one or two flights. This being the case we can conclude that it is possible to fly with this airline between any two of these five cities in, at most, two flights.

11. (a) The fact that there is symmetry across the leading diagonal tells us that there are no "one way only" connections.
 (b) Viewing the matrix $X + X^2 + X^3$, the presence of some zeros indicates that some hubs cannot be linked using 1, 2 or 3 direct links.
 Viewing the matrix $X + X^2 + X^3 + X^4$, the complete absence of zeros indicates that all hubs can be joined using 1, 2, 3 or 4 direct links.
 Thus all connections between two hubs can be done in up to four inter-hub connections, and some pairs do need four direct connections to connect them, for example A to J which could be linked using the path A-F-E-G-J.
 (c) Graph not shown here.

Miscellaneous Exercise Five. Page 148.

1. (a) Traversable. (b) Not traversable. (c) Traversable.
 (d) Traversable. (e) Traversable. (f) Not traversable.
 (g) Traversable. (h) Traversable.

Networks g and h are Eulerian.

Networks a, c, d and e are semi-Eulerian.

2. In a simple interest situation an initial amount, \$P, earns the same amount of interest, say \$I, each year. Thus, initial value = \$P
 Value after 1 year = \$P + \$I
 Value after 2 years = \$P + 2\$I
 Value after 3 years = \$P + 3\$I etc.

The constant increase from one term to the next is the characteristic of an arithmetic sequence.

The sequence of balances for simple interest is an example of an arithmetic sequence.

3. In a compound interest situation an initial amount, \$P, earns $r\%$ interest each year.
 Thus, initial value = \$P
 Value after 1 year = $\$P \times (1 + r/100)$
 Value after 2 years = $\$P \times (1 + r/100)^2$
 Value after 3 years = $\$P \times (1 + r/100)^3$ etc.

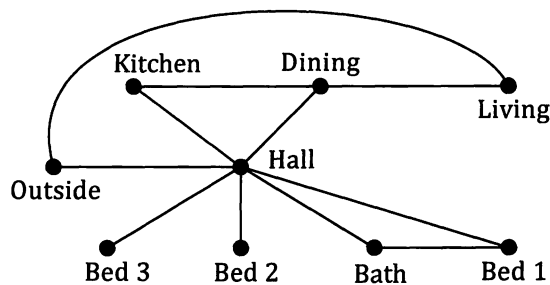
The constant multiplication from one term to the next is the characteristic of a geometric sequence.

The sequence of balances for compound interest is an example of a geometric sequence.

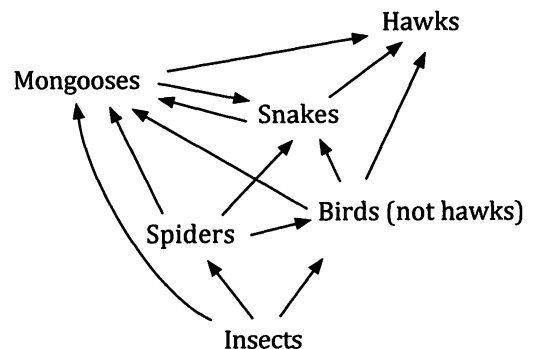
4. $T_1 = 5$, $T_2 = 17$, $T_3 = 53$, $T_4 = 161$, $T_5 = 485$.

5. $T_1 = 5$, $T_2 = 13$, $T_3 = 29$, $T_4 = 61$, $T_5 = 125$.

6.



7. Without knowing more about the particular species of spiders, birds, snakes etc that frequent the area it is difficult to be sure exactly who might eat who. Hence the food web shown on the right is just a possible suggestion. There are spiders that eat birds, for example, and some spiders can eat snakes, so although the suggested answer on the right does not include these two particular examples of "who is being eaten by who" it would not necessarily be wrong to include such directed edges.



8. (a) \$22000 (b) \$27000
 9. At the end of 32nd week the number of fish in the lake will fall to below 8000. Hence it will take 32 weeks (approximately, given that initial value of 10000 is only approximate and the 1.5% might vary somewhat).
 10. (a) 0.91, $S = 1.23P - 34.9$ (b) 193 cm (c) $P = 0.67S + 53.7$ (d) 178 cm
 11. Allow others in your class to read and comment upon your report and you read and comment on theirs.

Exercise 6A. Page 154.

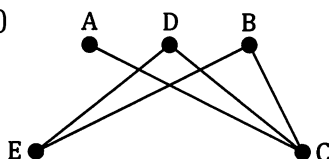
1. 105 units, HBW.
2. 105 units, HBW.
3. 95 units, HACW.
4. 90 units, ABEFZ.
5. 120 units, ACZ.
6. 46 units, ACGZ.
7. 125 units, ADFHJKMZ.
8. 125 units, ABCDFHIZ.
9. 60 minutes, ABCEGIK.
10. Ayesville - Buckeridge - Devizes - Ensmere - Hawton. \$27 000 000.
11. (a) Meckering - Cunderdin - Quairading - Corrigin - Bullaring - Wickepin. 187 km.
(b) Meckering - York - Beverley - Brookton - Pingelly - Narrogin - Wagin. 233 km.
12. The quickest journey from A to H takes 95 minutes and is along the route ACEFH.
13. Cataby - Badgingarra - Eneabba - Three Springs - Morawa. 214 km.
14. 100 units, A-B-C-E-F-I.
15. Current Position - Waterfall - Derelict Cottage - Tourist Office - Fire Lookout - Next Campsite.
4 h 50 mins (= 260 minutes + 30 minutes for lunch break).
Current Position - Waterfall - Derelict Cottage - Tourist Office - Parker's Lookout - Tourist Office - Fire Lookout - Next Campsite. 6 h 10 mins (= 340 minutes + 30 minutes for lunch break).
16. The chosen route should be MCFH for a total cost of \$50 000.
The requirement to go through junction S does change the minimum cost. The new route now needs to be MBSDH for a total cost of \$53 000.
17. (a) AFGK (b) ABEHIK (c) ABEIK

Miscellaneous Exercise Six. Page 161.

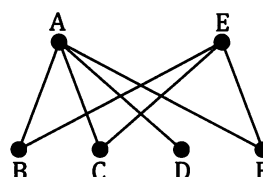
1.
$$\begin{matrix} & A & B & C & D \\ A & \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \\ B & \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \\ C & \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \\ D & \begin{bmatrix} 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$
2.
$$\begin{matrix} & A & B & C \\ A & \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \\ B & \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} \\ C & \begin{bmatrix} 1 & 2 & 0 \end{bmatrix} \end{matrix}$$
3.
$$\begin{matrix} & A & B & C & D & E \\ A & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \end{bmatrix} \\ B & \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \end{bmatrix} \\ C & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \end{bmatrix} \\ D & \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \end{bmatrix} \\ E & \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$
4.
$$\begin{matrix} & A & B & C & D \\ A & \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} \\ B & \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \\ C & \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix} \\ D & \begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$
5.
$$\begin{matrix} & A & B & C \\ A & \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \\ B & \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \\ C & \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \end{matrix}$$
6.
$$\begin{matrix} & A & B & C & D & E \\ A & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix} \\ B & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix} \\ C & \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \end{bmatrix} \\ D & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ E & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$
7. From the graph we can see that it is not possible to travel from one vertex to another (including back to itself) using exactly two edges. Hence the square of the adjacency matrix will be the 5×5 zero matrix.
8. (a) Head, tail, tail, head, head.
9. (a) The graph has exactly two odd vertices, vertex A (order 1) and vertex E (order 3). Thus whilst it is possible to find a trail that travels every edge once and once only, such a trail will start at one of the vertices A or E and finish at the other, for example the trail ABCDBEDE. The existence of such a trail, i.e. one that starts at one vertex, finishes at a different vertex, and travels along each and every edge once only makes the graph semi-Eulerian.
(b) If we add edge AE to the graph all vertices will be even. We can then find a trail that will start and finish at the same point and pass along each and every edge once only. For example ABCDBEDEA. The existence of such a closed trail means that the graph will then be Eulerian.
10. (a) ABDE, 12 units. ABDCE, 11 units. ABCDE, 19 units. ABCE, 12 units.
ACDE, 19 units. ACE, 12 units. ACBDE, 28 units.
(b) ABDCE.
11. CD, DE, GH, JK, KL, ML.
12. There are two equally shortest paths, each of length 23 units: ABCDFGH. ABCDEH
13. (a) Approximately 81%. ($r^2 = 0.8098$) (b) Approximately 24%. ($r^2 = 0.2419$)
14. $T_1 = 4$, $T_{n+1} = T_n + 1$.
15. (a) 7 (b) 17 (c) 2560 (d) 3584

16. Only (b) and (c):

(b)



(c)



17. (a) 0.85 (b) $M = 0.95E - 11.07$ (c) 51
 18. Cheapest upgrade will cost \$144 000 and will be ACDEHJLZ.
 19. (a) All three are connected graphs.
 (b) Only the middle of the three (the tetrahedron) is a complete graph.
 (c) Only the last one, the octahedron, is an Eulerian graph.
 (d) All three graphs are Hamiltonian graphs.
 20. 100 minutes. Passing through Barton, Dane, Grange.

	A	B	C	D	E	Totals
Stack College	77	93	60	9	2	241
Bosun College	24	52	98	14	8	196
Totals	101	145	158	23	10	437

Examining row percentages:

	A	B	C	D	E	Totals
Stack College	32%	39%	25%	4%	1%	101%*
Bosun College	12%	27%	50%	7%	4%	100%

* Rounding anomaly.

As we move from one College's results to the other the proportion of students achieving each grade changes markedly. 32%, almost one third, of all the Stack College students achieved an A grade compared with 12%, about one eighth, of the Bosun College students. The proportions achieving B grades were also strongly in favour of Stack College, 39%, compared to Bosun College, 27%. The changes in proportions achieving each grade as we move from the results of the students from one college to the other indicate that there is an association between the grades achieved and the college attended.

22. (a) There are six Hamiltonian circuits that can start at A and finish at A, though these really are three pairs of circuits with each pair consisting of the same circuit travelled in opposite directions.

The Hamiltonian circuits are:

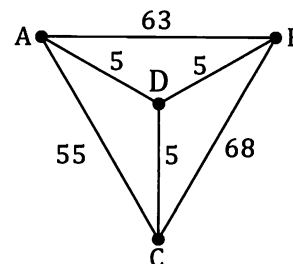
ACBDA and ADBCA each with a total estimated time of 259 minutes.

ABCD A and ADCBA each with a total estimated time of 246 minutes.

ABDCA and ACDBA each with a total estimated time of 241 minutes.

Thus the Hamiltonian circuit given in the question, ACBDA, is not the quickest. The circuit ABDCA, travelled as stated or as ACDBA, gives the quickest time of 241 minutes.

- (b) If we are only at A at start and finish and in between we visit each of B, C and D exactly once then it will be a Hamiltonian circuit that gives the shortest time. However, if multiple visits to the other vertices (and repeated trips along edges) are allowed it could be that the times make a non Hamiltonian path quicker. For the times shown on the right for example, and with repeat visits to any of B, C and D allowed, the quickest journey would be to avoid the three time consuming edges and follow ADBDCDA (or ADCDBDA) for a total of 30 minutes.



23. The withdrawal of \$55 000 can last for ten years. At the end of the eleventh year the final withdrawal would be \$16916 (nearest dollar).
 24. Allow others in your class to read and comment on your report and you read and comment on theirs.